

Fusions of One-Variable First-Order Modal Logics

Roman Kontchakov

Birkbeck, University of London, UK
r.kontchakov@bbk.ac.uk

Dmitry Shkatov

University of the Witwatersrand,
Johannesburg, South Africa
dmitry.shkatov@wits.ac.za

Frank Wolter

University of Liverpool, UK
wolter@liverpool.ac.uk

We investigate preservation results for the independent fusion of one-variable first-order modal logics. We show that, without equality, Kripke completeness and decidability of global and local consequence relations are preserved, under both expanding and constant domain semantics. By contrast, Kripke completeness and decidability are not preserved for fusions with equality and non-rigid constants (or, equivalently, counting up to one), again for the global and local consequence and under both expanding and constant domain semantics. This result is shown by encoding Diophantine equations. Even without equality, the finite model property is preserved only in the local case. Finally, we view fusions of one-variable modal logics as fusions of propositional modal logics sharing an **S5** modality and provide a general sufficient condition for transfer of Kripke completeness and decidability (but not of finite model property).

1 Introduction

The study of combined modal logics has been an active area of research for many years [10, 23, 35]. Only a few methods of combining modal systems have proved to be robust in the sense that key properties of the component logics are always inherited by the resulting combined logic. A central example of such a robust combination is fusion: a basic method in which the component logics are combined without introducing axioms that mix their modal operators, and, correspondingly, without imposing any semantic constraints that relate the distinct accessibility relations. In fact, properties such as Kripke completeness, decidability, finite model property, and Craig interpolation are all preserved under forming the fusion of two (or more) classical propositional modal logics, for both the local and global consequence relations [31, 21, 9, 33, 4, 14, 2, 6].

The aim of this contribution is to initiate a systematic investigation into the extent to which these preservation results for fusions of propositional modal logics can be lifted to (fragments of) first-order modal logics and corresponding many-dimensional modal logics. We focus in this paper on one-variable first-order modal logics, either without equality or with equality and non-rigid constants, and on corresponding commutators and products of propositional modal logics. These logics have been studied extensively [11, 19, 24, 5, 28] as they are often computationally and semantically better behaved than full first-order modal logics. For fusions without equality our main result is as follows:

Theorem A (Equality-Free Fusion). *Kripke completeness and decidability are preserved under fusions of Kripke complete one-variable first-order modal logics, for both the local and global consequence and under both expanding and constant domain semantics. The finite model property is preserved for the local consequence but not for the global consequence. Moreover, the finite model property fails for the global consequence for all non-trivial fusions under both expanding and constant domain semantics.*

The proof builds on the cactus model construction [16], extending it to one-variable modal logic via the quasimodel technique [25]. For one-variable first-order modal logics with equality, the presence of equality can only have a substantial impact if one also allows non-rigid constants or, equivalently, counting quantifiers such as $\exists^=1 x P(x) \equiv \exists x [P(x) \wedge \forall y (P(y) \rightarrow (x = y))]$, which expresses that

predicate P is interpreted as a singleton. In this setting, drawing on known decidability results for the one-variable first-order extension of the modal logic of inequality (also known as the ‘elsewhere’ logic) and for one-variable temporal logics [19, 17, 12] and by encoding Diophantine equations and Minsky machines to show undecidability, we establish the following non-transfer result:

Theorem B (Fusion with Equality). *Decidability and recursive axiomatisability are not preserved for fusions of one-variable first-order modal logics with equality, for both the local and global consequence and under both expanding and constant domain semantics. Moreover, the global consequence is undecidable for all non-trivial fusions for constant domain semantics.*

We stress that it would be a mistake to infer that fusions of one-variable modal logics with equality are broadly undecidable. In fact, when based on standard modal systems such as polymodal **K** or **S5**, one-variable modal logics with equality remain decidable, even in CONEXPTIME [1].

Since the one-variable fragment of first-order logic without equality can be identified with propositional modal logic **S5**, one-variable first-order modal logics can often be viewed as propositional multimodal logics equipped with an **S5** modality. Working within this perspective, we provide a sufficient condition for the transfer of Kripke completeness and decidability in fusions of propositional modal logics that share an **S5** modality. We say that a propositional modal logic admits E -homogeneous models for an equivalence relation E interpreting an **S5** modality $\Box_E$ if, for sufficiently large infinite cardinals κ , it is complete for Kripke models in which, for every E -equivalence class W of worlds, each formula is either not satisfied anywhere in W or is satisfied in κ -many worlds of W .

Theorem C (Propositional Fusion with Shared **S5**). *Kripke completeness and decidability are preserved for fusions of propositional modal logics admitting E -homogeneous models for a shared **S5** modality $\Box_E$, for both the local and global consequence.*

The conditions of Theorem C hold, for example, for Kripke complete (semi)commutators with **S5** and for (expanding) product modal logics with **S5**. Hence, it provides transfer results for corresponding one-variable modal logics. We note that, in contrast to the proof of Theorem A that uses quasimodels, one cannot directly prove transfer of the finite model property based on homogeneous models. Theorem C addresses a problem posed by Baader, Ghilardi and Tinelli [2].

2 Preliminaries

We consider first-order (multi-)modal logics in languages with a single, fixed, individual variable. A language $\mathcal{L}_M$ contains countably many unary predicate letters, the individual variable x , the Booleans $\neg$ and $\wedge$, the quantifier symbol $\exists$, and a finite set M of unary (necessity-like) modal operators, denoted by $\Box$, possibly with subscripts. Thus, in $\mathcal{L}_M$, atomic formulas have the form $P(x)$, where P is a predicate letter. Formulas of $\mathcal{L}_M$, denoted by $\varphi, \psi, \dots$, are built from atomic ones in the usual way, the only allowed quantifier being $\exists x$. The symbols $\perp, \rightarrow, \vee, \forall$, and $\Diamond$ are standard abbreviations. We also denote $\Box^{\leq n} \varphi = \varphi \wedge \Box \varphi \wedge \dots \wedge \Box^n \varphi$. Proposition letters $p, q, \dots$ are abbreviations for $\exists x P(x), \exists x Q(x)$, etc. We also consider languages with equality: $\mathcal{L}_M^=$ contains, in addition to the symbols of $\mathcal{L}_M$, the binary predicate symbol $=$ and constant symbols, denoted by $c_1, c_2, \dots$. Thus, in $\mathcal{L}_M^=$, a term t is either x or a constant c_i , and atomic formulas have the form $P(t)$ or $t_1 = t_2$. We usually identify languages with sets of their formulas. The set of subformulas of a formula φ is denoted by $\text{sub } \varphi$.

By a *one-variable (first-order) M -logic*, we mean a set of $\mathcal{L}_M$ -formulas that includes the one-variable fragment of the classical first-order logic **FO** and formulas $\Box(P(x) \rightarrow Q(x)) \rightarrow (\Box P(x) \rightarrow \Box Q(x))$, for all $\Box \in M$, and is closed under Modus Ponens (MP), Generalisation ($\varphi / \forall x \varphi$), Necessitation ($\varphi / \Box \varphi$, for

each $\Box \in M$), and Substitution (of arbitrary one-variable formulas for formulas of the form $P(x)$).¹ If L is a one-variable M -logic and $\Gamma \subseteq \mathcal{L}_M$, then $L + \Gamma$ denotes the smallest one-variable M -logic including $L \cup \Gamma$; we write $L + \varphi$ for $L + \{\varphi\}$. We note that every one-variable M -logic contains the converse Barcan formula $CBF_\Box = \Box \forall x P(x) \rightarrow \forall x \Box P(x)$, for each $\Box \in M$.

Let L be a one-variable M -logic and $\varphi, \psi \in \mathcal{L}_M$. We write $\varphi \vdash_L \psi$ and say that ψ is a *local consequence* of φ in L if ψ belongs to the smallest set including $L \cup \{\varphi\}$ and closed under MP and Generalisation. By the deduction theorem, L is decidable iff $\vdash_L$ is decidable. We write $\varphi \vdash_L^* \psi$ and say that ψ is a *global consequence* of φ in L if ψ belongs to the smallest set including $L \cup \{\varphi\}$ and closed under MP, Generalisation, and Necessitation for all $\Box \in M$. We call L *globally decidable* if the global consequence in L is decidable.

The *fusion* of a one-variable M_1 -logic L_1 and a one-variable M_2 -logic L_2 , where $M_1 \cap M_2 = \emptyset$, is the smallest one-variable $(M_1 \cup M_2)$ -logic $L_1 \otimes L_2$ that includes $L_1 \cup L_2$.

We are mostly concerned with logics defined in terms of Kripke semantics. A *Kripke M -frame* is a tuple $\mathfrak{F} = (W, \{R_\Box\}_{\Box \in M})$, where W is a non-empty set of *worlds* and $R_\Box \subseteq W \times W$, for all $\Box \in M$. We write $R_\Box(w)$ for $\{w' \in W \mid wR_\Box w'\}$. We say that $\mathfrak{F}$ is *non-trivial* if there are distinct $w, v \in W$ with $wR_\Box v$, for some $\Box \in M$. A *Kripke cd -model* based on $\mathfrak{F}$ is a pair $\mathfrak{M} = (\mathfrak{F}, I)$, where $I = \{\langle \Delta, I_w \rangle \mid w \in W\}$ is a family of classical first-order models. A *Kripke xd -model* based on $\mathfrak{F}$ is a pair $\mathfrak{M} = (\mathfrak{F}, I)$, where $I = \{\langle \Delta_w, I_w \rangle \mid w \in W\}$ is a family of classical first-order models satisfying the *expanding domains condition*: $\Delta_w \subseteq \Delta_v$, for all $w, v \in W$ with $wR_\Box v$ and all $\Box \in M$. For languages with equality and constants, the classical models associated with worlds are required to be *normal* (i.e., $=$ is interpreted as identity); the resultant Kripke models are called *Kripke d -models with equality*, for $d \in \{xd, cd\}$; note that rigid interpretation of constants, i.e., $I_w(c_k) = I_v(c_k)$, for all $w, v \in W$, is *not* assumed. We say that a Kripke model is *finite* if its set of worlds is finite and every classical model associated with its worlds is finite.

Since cd -models are a special case of xd -models, we define truth and validity only for the latter. Let $\mathfrak{M} = (W, \{R_\Box\}_{\Box \in M}, \langle \Delta_w, I_w \rangle_{w \in W})$ be an xd -model. An *assignment* in $\mathfrak{M}$ is a map α sending x to an element of the set $\bigcup_{w \in W} \Delta_w$. The *truth* of one-variable $\mathcal{L}_M$ - and $\mathcal{L}_M^-$ -formulas at world w of $\mathfrak{M}$ under assignment α is defined by recursion on the formula construction in the standard way, in particular:

- $\mathfrak{M}, w \models^\alpha \exists x \varphi$ if $\mathfrak{M}, w \models^{\alpha'} \varphi$, for some α' such that $\alpha'(x) \in \Delta_w$;
- $\mathfrak{M}, w \models^\alpha t_1 = t_2$ if $|t_1|_w^\alpha = |t_2|_w^\alpha$, where $|t|_w^\alpha$ denotes the value of the term t at w under α .

We say that φ is *true at $w \in W$* , and write $\mathfrak{M}, w \models \varphi$, if $\mathfrak{M}, w \models^\alpha \varphi$ holds for every α with $\alpha(x) \in \Delta_w$. We also say that φ is *true in $\mathfrak{M}$* , and write $\mathfrak{M} \models \varphi$, if $\mathfrak{M}, w \models \varphi$ holds for every $w \in W$.

Let $d \in \{xd, cd\}$. We write $\mathfrak{F} \models_d \varphi$ if φ is true in every d -model based on $\mathfrak{F}$. Then $\mathfrak{F}$ is a *d -frame* for a logic L , in symbols $\mathfrak{F} \models_d L$, if $\mathfrak{F} \models_d \varphi$ for all $\varphi \in L$. An $\mathcal{L}_M$ -formula φ is *d -valid* on a class $\mathcal{C}$ of Kripke M -frames (written as $\mathcal{C} \models_d \varphi$) if $\mathfrak{F} \models_d \varphi$, for all $\mathfrak{F} \in \mathcal{C}$. Analogous notions are defined for formulas and models with equality. We define the following sets of formulas:

$$\text{Log}_d \mathcal{C} = \{\varphi \in \mathcal{L}_M \mid \mathcal{C} \models_d \varphi\} \quad \text{and} \quad \text{Log}_d^- \mathcal{C} = \{\varphi \in \mathcal{L}_M^- \mid \mathcal{C} \models_d \varphi\}.$$

Note that, for every $\mathcal{C}$, we have $\mathcal{C} \models_{cd} \forall x \Box P(x) \rightarrow \Box \forall x P(x)$ (the Barcan formula, $BF_\Box$), for all $\Box \in M$.

Let L be a one-variable M -logic, $\mathcal{C}$ a class of M -frames, and $d \in \{xd, cd\}$. We say L is *d -complete* for $\mathcal{C}$ if $L = \text{Log}_d \mathcal{C}$, and that L is *globally d -complete* for $\mathcal{C}$ in case $\varphi \vdash_L^* \psi$ iff $\mathfrak{M} \models \varphi$ implies $\mathfrak{M} \models \psi$, for every d -model $\mathfrak{M}$ based on a frame from $\mathcal{C}$. Note that global d -completeness implies d -completeness,

¹One-variable fragments of first-order modal logics are one-variable logics in our sense, but not every one-variable logic is a one-variable fragment of a first-order modal logic; see, e.g., [29, Section 3].

but the converse is not always true. We say that L has the *finite d-model property* (*d-fmp*) if, for every $\varphi \notin L$, there exists a finite d -model $\mathfrak{M}$ such that $\mathfrak{M} \models L$ and $\mathfrak{M} \not\models \varphi$. We say that L has the *global finite d-model property* (*global d-fmp*) if, for every φ and ψ with $\varphi \not\vdash_L^* \psi$, there exists a finite d -model $\mathfrak{M}$ such that $\mathfrak{M} \models L \cup \{\varphi\}$ and $\mathfrak{M} \not\models \psi$.

The *fusion* of Kripke frames $\mathfrak{F}_1 = (W, \{R_\square\}_{\square \in M_1})$ and $\mathfrak{F}_2 = (W, \{R_\square\}_{\square \in M_2})$, where $M_1 \cap M_2 = \emptyset$, is the frame $\mathfrak{F}_1 \otimes \mathfrak{F}_2 = (W, \{R_\square\}_{\square \in M_1 \cup M_2})$. If $\mathcal{C}_1$ and $\mathcal{C}_2$ are classes of Kripke frames, then $\mathcal{C}_1 \otimes \mathcal{C}_2$ denotes the class of frames comprising $\mathfrak{F}_1 \otimes \mathfrak{F}_2$, for all $\mathfrak{F}_1 \in \mathcal{C}_1$ and $\mathfrak{F}_2 \in \mathcal{C}_2$ with the same set of worlds.

We will also consider propositional modal logics that can be viewed as fragments of the one-variable modal logics introduced above. Our notation and definitions are standard, following the conventions established earlier; see also the full version [20]. The *fusion* of a propositional M_1 -logic L_1 and a propositional M_2 -logic L_2 , where $M_1 \cap M_2 = \emptyset$, is the smallest propositional $(M_1 \cup M_2)$ -logic $L_1 \otimes L_2$ that includes $L_1 \cup L_2$. The *semicommutator*² and the *commutator* [25, 23] of a propositional M -logic L with the propositional monomodal logic **S5** are the propositional $(M \cup \{\Box_E\})$ -logics

$$[L, \mathbf{S5}]^{xd} = L \otimes \mathbf{S5} + \{(lcom_\square) \mid \square \in M\} \quad \text{and} \quad [L, \mathbf{S5}]^{cd} = [L, \mathbf{S5}]^{xd} + \{(rcom_\square) \mid \square \in M\},$$

where $(lcom_\square)$ and $(rcom_\square)$ are the following formulas with their Kripke frame correspondents:

$$\begin{aligned} \Box \Box_E p &\rightarrow \Box_E \Box p, & E \circ R &\subseteq R \circ E, & (lcom_\square) \\ \Box_E \Box p &\rightarrow \Box \Box_E p, & R \circ E &\subseteq E \circ R. & (rcom_\square) \end{aligned}$$

There exists a well-known translation of one-variable M -formulas into propositional $(M \cup \{\Box_E\})$ -formulas: $P(x)^* = p$; $(\neg \varphi)^* = \neg \varphi^*$; $(\varphi_1 \wedge \varphi_2)^* = \varphi_1^* \wedge \varphi_2^*$; $(\Box \varphi)^* = \Box \varphi^*$, for all $\Box \in M$; $(\exists x \varphi)^* = \Diamond_E \varphi^*$. We note that $CBF_\square^* = (lcom_\square)$ and $BF_\square^* = (rcom_\square)$. If $\Gamma \subseteq \mathcal{L}_M$, then $\Gamma^* = \{\varphi^* \mid \varphi \in \Gamma\}$.

If L is a propositional M -logic, then the smallest one-variable M -logic containing all the substitution instances of formulas from L (here, we substitute arbitrary one-variable formulas for proposition letters) is denoted by $\mathcal{Q}L$.

Lemma 2.1. *If L is a propositional M -logic, then $\mathcal{Q}L^* = [L, \mathbf{S5}]^{xd}$ and $(\mathcal{Q}L + BF)^* = [L, \mathbf{S5}]^{cd}$.*

Proof. Let $L' = [L, \mathbf{S5}]^{xd}$ and let $\mathbf{FO}_1$ be the one-variable fragment of the classical first-order logic. It is easy to check that $\vdash_{L'} \Diamond \Box_E p \rightarrow \Box_E \Diamond p (= chr_\Diamond)$, for each $\Diamond$ corresponding to $\Box \in M$. We also rely on the well-known [32] fact that $\mathbf{FO}_1^* = \mathbf{S5}$.

First, we prove that the translation $\cdot^*$ commutes with substitution: for all one-variable formulas $\varphi, \psi_1, \dots, \psi_n$,

$$(\varphi)^*[\psi_1, \dots, \psi_n/P_1(x), \dots, P_n(x)] = \varphi^*[\psi_1^*, \dots, \psi_n^*/p_1, \dots, p_n]. \quad (*)$$

The proof is by induction on φ ; we only consider the case $\varphi = \exists x \theta$. Let $\sigma = [\psi_1, \dots, \psi_n/P_1(x), \dots, P_n(x)]$ and $\sigma^* = [\psi_1^*, \dots, \psi_n^*/p_1, \dots, p_n]$. Since, by IH, $(\theta \sigma)^* = \theta^* \sigma^*$, it follows that

$$((\exists x \theta) \sigma)^* = (\exists x (\theta \sigma))^* = \Diamond_E (\theta \sigma)^* = \Diamond_E (\theta^* \sigma^*) = (\Diamond_E \theta^*) \sigma^* = (\exists x \theta)^* \sigma^*.$$

We now prove that $L' \subseteq \mathcal{Q}L^*$. It should be clear that $L \subseteq \mathcal{Q}L^*$, and that $\mathbf{S5} = \mathbf{FO}_1^* \subseteq \mathcal{Q}L^*$. Since $CBF_\square$ belongs to every one-variable modal logic and $CBF_\square^* = (lcom_\square)$, it follows that $(lcom_\square) \in \mathcal{Q}L^*$. Now, the claim follows from $(*)$ and the closure of $\mathcal{Q}L^*$ under MP, Necessitation, and Generalisation.

²In general, semicommutators are also required to contain confluence axioms for commuting modalities; these are derivable in semicommutators with **S5**.

We next prove that $\mathcal{Q}L^* \subseteq L'$. Here, we use the fact that $\mathbf{FO}_1$ is obtained by adding to the classical propositional logic the axiom $\forall x P(x) \rightarrow P(x)$ and the rule $\psi \rightarrow \varphi(x) / \psi \rightarrow \forall x \varphi(x)$, where x is not free in ψ . In view of (*), the only non-trivial part is to show that $\vdash_{L'} (\psi \rightarrow \varphi)^*$ implies $\vdash_{L'} (\psi \rightarrow \forall x \varphi)^*$ provided that x is not free in ψ . To this end, we show that, if γ is a propositional $(M \cup \{\Box_E\})$ -formula and β is a propositional $(M \cup \{\Box_E\})$ -formula such that all occurrences of proposition letters occur within the scope of $\Box_E$, then $\vdash_{L'} \beta \rightarrow \gamma$ implies $\vdash_{L'} \beta \rightarrow \Box_E \gamma$. Since $\Box_E$ is monotonic, it is enough to show that $\vdash_{L'} \beta \rightarrow \Box_E \beta$, for all β of the assumed form.

Due to our assumption about its form, we may rewrite β with $\neg$, $\wedge$, $\vee$, $\Box_E$, $\Box$, and $\Diamond$ so that all negations occur either within the scope of $\Box_E$ or immediately precede $\Box_E$ (to that end, we push all the negations outside of any $\Box_E$ down until they reach some $\Box_E$). Thus, β is built from formulas of the form $\Box_E \delta$ and $\neg \Box_E \delta$ using $\wedge$, $\vee$, $\Box$, and $\Diamond$. We proceed by induction on the structure of such β .

- If $\beta = \Box_E \gamma$ or $\beta = \neg \Box_E \gamma$, then the claim holds because $\Box_E$ is an **S5** modality, and therefore $\vdash_{L'} \Box_E \gamma \rightarrow \Box_E \Box_E \gamma$ and $\vdash_{L'} \neg \Box_E \gamma \rightarrow \Box_E \neg \Box_E \gamma$.
- The cases of $\wedge$ and $\vee$ are straightforward.
- If $\beta = \Box \gamma$ and $\vdash_{L'} \gamma \rightarrow \Box_E \gamma$, then, by monotonicity of $\Box$ and $(lcom_\Box)$, $\vdash_{L'} \Box \gamma \rightarrow \Box_E \Box \gamma$.
- If $\beta = \Diamond \gamma$ and $\vdash_{L'} \gamma \rightarrow \Box_E \gamma$, then, by monotonicity of $\Diamond$ and $(chr_\Box)$, $\vdash_{L'} \Diamond \gamma \rightarrow \Box_E \Diamond \gamma$.

To prove the claim of the lemma for $[L, \mathbf{S5}]^{cd}$, it suffices to recall that $BF_\Box^* = (rcom_\Box)$. $\square$

3 Fusions of Equality-Free One-Variable Logics

In this section, we prove a precise version of Theorem A.

Theorem 3.1. *Let $\mathcal{C}_1, \mathcal{C}_2$ be non-empty classes of frames closed under disjoint unions, $d \in \{xd, cd\}$, and L_i be one-variable modal logics (globally) d -complete for $\mathcal{C}_i$, for both $i = 1, 2$. Then*

- $L_1 \otimes L_2$ is (globally) d -complete for $\mathcal{C}_1 \otimes \mathcal{C}_2$;
- $L_1 \otimes L_2$ is (globally) decidable if both L_1 and L_2 are (globally) decidable;
- $L_1 \otimes L_2$ has the d -fmp if both L_1 and L_2 have the d -fmp.

If $\mathcal{C}_1$ and $\mathcal{C}_2$ contain non-trivial frames, then $L_1 \otimes L_2$ does not have the global d -fmp.

We begin the proof of these results by defining a general structure of frames for fusions, which follows Goranko and Passy's cactus model construction [16].

Definition 3.2. An *i-cactus frame* (W, R_1, R_2) with thorns $V \subseteq W$ is defined inductively.

basis A 1-frame $(W^0, R_1^0) \in \mathcal{C}_1$ gives rise to a 2-cactus frame $(W^0, R_1^0, \emptyset)$ with thorns W^0 . A 1-cactus frame is defined symmetrically.

induction step Let (W, R_1, R_2) be a 2-cactus frame with thorns V and $(W^v, R_2^v) \in \mathcal{C}_2$, for $v \in V$, a collection of 2-frames such that, for each $v \in V$, the following conditions hold:

$$W \cap W^v = \{v\} \quad \text{and} \quad W^v \cap W^{v'} = \emptyset, \text{ for all } v' \in V \text{ with } v' \neq v. \quad (1)$$

Then $(W', R_1, R_2 \cup \bigcup_{v \in V} R_2^v)$ is a 1-cactus frame with thorns $W' \setminus W$, where $W' = W \cup \bigcup_{v \in V} W^v$.

Given a 1-cactus frame and a collection of 1-frames, a 2-cactus frame is defined symmetrically.

A *limit i-cactus frame* is obtained as the limit of the sequence of cactus frames starting from an *i*-frame.

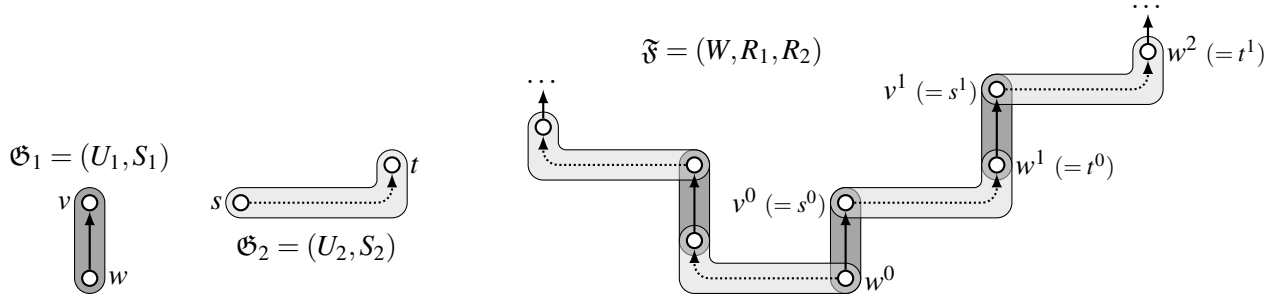


Figure 1: Sample non-trivial frames $\mathfrak{G}_1$ and $\mathfrak{G}_2$ and resulting limit 1-cactus frame $\mathfrak{F}$: solid arrows with dark background for R_1/S_1 -components and dotted arrows with light background for R_2/S_2 -components.

The following property of limit i -cactus frames is immediate from (1):

Proposition 3.3. *A limit i -cactus frame is a frame from $\mathcal{C}_1 \otimes \mathcal{C}_2$.*

With these definition and property at hand, we can establish the general non-transfer of the global d -fmp claimed in Theorem 3.1.

Lemma 3.4. *Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be classes of frames closed under disjoint unions containing non-trivial frames, $d \in \{xd, cd\}$, and $L_i = \text{Log}_d \mathcal{C}_i$, for $i = 1, 2$. Then $L_1 \otimes L_2$ does not enjoy the global d -fmp.*

Proof. We define the modal operators

$$\Diamond_p \psi = \Diamond_1(\neg p \wedge \Diamond_2(p \wedge \psi)) \quad \text{and} \quad \Box_p \psi = \neg \Diamond_p \neg \psi,$$

where p is a proposition letter not in ψ , and consider the formula (‘every dog has its day’)

$$\varphi = p \rightarrow [\exists x (\neg Q(x) \wedge \Diamond_p Q(x)) \wedge \forall x (Q(x) \rightarrow \Box_p Q(x))].$$

We show that $\varphi \not\models_{L_1 \otimes L_2}^* \neg p$ and that this cannot be witnessed by a finite model.

First, we note that, if $\mathfrak{M}$ is a model based on a frame $\mathfrak{F} = (W, R_1, R_2)$ for $L_1 \otimes L_2$ and $w^0 \in W$ such that $\mathfrak{M}, w^0 \models p$ and $\mathfrak{M} \models \varphi$, then $\mathfrak{M}$ is infinite: such a model $\mathfrak{M}$ contains an infinite $R_1 R_2$ -path of worlds, starting at w^0 , with the ever-increasing, non-universal extension of Q , all of which make p true.

Next, to show that $\varphi \not\models_{L_1 \otimes L_2}^* \neg p$, we define an $L_1 \otimes L_2$ -model $\mathfrak{M}$ such that $\mathfrak{M} \models \varphi$, but $\mathfrak{M}, w^0 \not\models p$, for some world w^0 . Let $\mathfrak{G}_1 = (U_1, S_1)$ and $\mathfrak{G}_2 = (U_2, S_2)$ be non-trivial frames from, respectively, $\mathcal{C}_1$ and $\mathcal{C}_2$. Thus, $\mathfrak{G}_1$ contains worlds w and v such that $w \neq v$ and $w S_1 v$; similarly, $\mathfrak{G}_2$ contains worlds s and t such $s \neq t$ and $s S_2 t$; see Fig. 1 for a simple example. Let $\mathfrak{F} = (W, R_1, R_2)$ be the limit 1-cactus frame constructed from the sequence $\mathfrak{G}_1^i = (U_1^i, S_1^i)$, $i \in \mathbb{N}$, of disjoint copies of $\mathfrak{G}_1$ and the sequence $\mathfrak{G}_2^i = (U_2^i, S_2^i)$, $i \in \mathbb{N}$, of disjoint copies of $\mathfrak{G}_2$, starting with $\mathfrak{G}_1^0$; we identify v^i with s^i and t^i with w^{i+1} , for all $i \in \mathbb{N}$, where u^i denotes the copy of u from U_k^i , for $i \in \mathbb{N}$. By Proposition 3.3, $\mathfrak{F} \in \mathcal{C}_1 \otimes \mathcal{C}_2$, and so $\mathfrak{F}$ is a frame for $L_1 \otimes L_2$. Let $W^0 = \{w^0\}$ and, for all $i \in \mathbb{N}$, let $V^i = R_2(W^i)$ and $W^{i+1} = R_1(V^i)$. Now, define a model $\mathfrak{M} = (\mathfrak{F}, \langle \mathbb{N}, I_w \rangle_{w \in W})$ by putting $I_w(Q) = \{j \in \mathbb{N} \mid j < i\}$ if $w \in W^i$, and $I_w(Q) = \emptyset$ otherwise, and by making p true at $w \in W$ iff $w \in \bigcup_{i \in \mathbb{N}} W^i$. Then $\mathfrak{M} \models \varphi$ but $\mathfrak{M}, w^0 \not\models \neg p$. $\square$

The proofs of the remaining transfer results claimed in Theorem 3.1 lift the cactus model construction to the one-variable modal logic using the quasimodel technique; see, e.g., [25]. We require two additional main ingredients: (a) types and quasistates for a finite representation of any number of domain elements and (b) surrogates for separating the two components of the fusion.

In the sequel, we denote the language of $L_1 \otimes L_2$ by $\mathcal{L}$ and the language of L_i by $\mathcal{L}_i$, for $i = 1, 2$. For simplicity, we assume that L_1 and L_2 are monomodal logics with modalities $\Box_1$ and $\Box_2$, respectively. Let Φ be a finite set of $\mathcal{L}$ -formulas closed under subformulas. A Φ -type t is a maximal Boolean-consistent subset of $\{\psi, \neg\psi \mid \psi \in \Phi\}$, that is, $\neg\psi \in t$ iff $\psi \notin t$, for each $\psi \in \Phi$, and $\psi_1 \wedge \psi_2 \in t$ iff $\psi_1, \psi_2 \in t$, for each $\psi_1 \wedge \psi_2 \in \Phi$. The number of Φ -types is bounded by $2^{|\Phi|}$ (for $\neg\psi \in \Phi$, a type contains either $\neg\psi$ or both $\neg\neg\psi$ and ψ). A Φ -quasistate is a non-empty set q of Φ -types such that

$$\exists x \psi \in t \quad \text{iff} \quad \text{there is } t' \in q \text{ with } \psi \in t', \quad \text{for each } \exists x \psi \in \Phi \text{ and } t \in q.$$

With a Φ -quasistate q , we associate the following *realisability* $\mathcal{L}$ -sentence:

$$\hat{q} = \forall x \bigvee_{t \in q} t(x) \wedge \bigwedge_{t \in q} \exists x t(x),$$

where $t(x)$ denotes the conjunction of formulas in t . If $\mathcal{Q}$ is a set of Φ -quasistates, then $\hat{\mathcal{Q}} = \bigvee \{\hat{q} \mid q \in \mathcal{Q}\}$.

Let φ be an $\mathcal{L}$ -formula. For each $\Box_i \psi \in \text{sub } \varphi$, we introduce a fresh unary predicate letter $P_{\Box_i \psi}$ and define the *surrogate* of $\Box_i \psi$ by taking $P_{\Box_i \psi}(x)$ if ψ has a free variable and $\forall x P_{\Box_i \psi}(x)$ if ψ is closed. Denote by φ^1 the $\mathcal{L}_1$ -formula obtained from φ by replacing $\Box_2 \psi$ not in the scope of another $\Box_2$ with its surrogate; the $\mathcal{L}_2$ -formula φ^2 is defined symmetrically.

As has already been observed in the propositional setting [21, 2], the analysis of local consequence is substantially more intricate than that of global consequence. We begin our proof with the latter case and show the following lemma which states transfer of *xd*-completeness of the global consequence and from which the transfer of decidability of global consequence follows by a simple enumeration of all possible Φ -quasistates.

Lemma 3.5. *Let $\mathcal{C}_1, \mathcal{C}_2$ be non-empty classes of frames closed under disjoint unions and L_i be one-variable modal logics globally *xd*-complete for $\mathcal{C}_i$, for both $i = 1, 2$. Let φ and ψ be $\mathcal{L}$ -formulas. Denote $\Phi = \text{sub } \varphi \cup \text{sub } \psi$. Then the following are equivalent:*

- (G1) $\varphi \Vdash_{L_1 \otimes L_2}^* \psi$;
- (G2) *there is an *xd*-model $\mathfrak{M}$ based on a frame from $\mathcal{C}_1 \otimes \mathcal{C}_2$ such that $\mathfrak{M} \models \varphi$ but $\mathfrak{M} \not\models \psi$;*
- (G3) *there is a set $\mathcal{Q}$ of Φ -quasistates such that*
 - (G3.1) $\varphi^1 \wedge \hat{\mathcal{Q}}^1 \Vdash_{L_1}^* \psi^1$,
 - (G3.2) $\varphi^1 \wedge \hat{\mathcal{Q}}^1 \Vdash_{L_1}^* \neg \hat{q}_i^1$, for each $q_i \in \mathcal{Q}$,
 - (G3.3) $\hat{\mathcal{Q}}^2 \Vdash_{L_2}^* \neg \hat{q}_i^2$, for each $q_i \in \mathcal{Q}$.

Proof. Implication (G2) $\Rightarrow$ (G1) follows from soundness of $\vdash_{L_1 \otimes L_2}^*$.

(G1) $\Rightarrow$ (G3) Suppose $\varphi \Vdash_{L_1 \otimes L_2}^* \psi$ and consider the following set $\mathcal{Q}$ of Φ -quasistates:

$$\mathcal{Q} = \{q \text{ a } \Phi\text{-quasistate} \mid \varphi \Vdash_{L_1 \otimes L_2}^* \neg \hat{q}\}.$$

Clearly, $\bigvee_q \hat{q}$, for all Φ -quasistates q , is an $L_1 \otimes L_2$ -tautology, and so $\varphi \vdash_{L_1 \otimes L_2}^* \bigvee_q \hat{q}$. Observe that $\hat{\mathcal{Q}}$ consists of the disjuncts left after removing from $\bigvee_q \hat{q}$ all q with $\varphi \vdash_{L_1 \otimes L_2}^* \neg \hat{q}$. Thus, we obtain $\varphi \vdash_{L_1 \otimes L_2}^* \hat{\mathcal{Q}}$. Next, $\varphi \Vdash_{L_1 \otimes L_2}^* \psi$ implies $\varphi \wedge \hat{\mathcal{Q}} \Vdash_{L_1 \otimes L_2}^* \psi$. Moreover, by definition, we also have $\hat{\mathcal{Q}} \Vdash_{L_1 \otimes L_2}^* \neg \hat{q}_i$ and therefore $\varphi \wedge \hat{\mathcal{Q}} \Vdash_{L_1 \otimes L_2}^* \neg \hat{q}_i$, for every $q_i \in \mathcal{Q}$. Then (G3.1)–(G3.3) follow by contraposition: for example, if we assume $\varphi \wedge \hat{\mathcal{Q}} \Vdash_{L_1 \otimes L_2}^* \psi$ but $\varphi^1 \wedge \hat{\mathcal{Q}}^1 \not\vdash_{L_1}^* \psi^1$, then the latter also provides an inference in $L_1 \otimes L_2$, whence, as a substitution instance, we obtain $\varphi \wedge \hat{\mathcal{Q}} \vdash_{L_1 \otimes L_2}^* \psi$ contrary to our assumption.

(G3) $\Rightarrow$ (G2) Suppose there is a set $\mathcal{Q}$ of Φ -quasistates such that (G3.1)–(G3.3) hold. To describe the construction of the witnessing *xd*-model $\mathfrak{M}$, we require the following set of definitions.

Definition 3.6. A Φ -run r through a frame (W, R_i) is a function from an R_i -closed subset of W , denoted by $\text{dom } r$, to Φ -types. For $w \in \text{dom } r$, a run r is called

- *i-coherent at w* if $\Box_i \psi \in r(w)$ implies $\psi \in r(w')$, for all $w' \in R_i(w)$;
- *i-saturated at w* if $\Box_i \psi \in r(w)$ whenever $\Box_i \psi \in \Phi$ and $\psi \in r(w')$, for all $w' \in R_i(w)$.

We say that r is *i-coherent on $W' \subseteq W$* if r is *i-coherent at each $w \in \text{dom } r \cap W'$* and *i-coherent* if r is *i-coherent on W* ; similarly for r being *i-saturated*.

An *i-quasimodel* $\mathfrak{Q}$ for Φ is a tuple of the form $((W, R_i), w_0, \mathfrak{R})$, where $(W, R_i) \in \mathcal{C}_i$, $w_0 \in W$ and $\mathfrak{R}$ is a set of Φ -runs through (W, R_i) such that each $r \in \mathfrak{R}$ is *i-coherent* and *i-saturated*, and each $\mathfrak{R}(w)$ is a Φ -quasistate, for $w \in W$, where we denote $\mathfrak{R}_w = \{r \in \mathfrak{R} \mid w \in \text{dom } r\}$ and $\mathfrak{R}(w) = \{r(w) \mid r \in \mathfrak{R}_w\}$.

Definition 3.7. An *i-cactus* $((W, R_1, R_2), w_0, \mathfrak{R})$ for Φ with thorns $V \subseteq W$ is defined inductively.

basis A 1-quasimodel $((W^0, R_1^0), w_0, \mathfrak{R}^0)$ for Φ with a 1-frame $(W^0, R_1^0) \in \mathcal{C}_1$ gives rise to a 2-cactus $((W^0, R_1^0, \emptyset), w_0, \mathfrak{R}^0)$ for Φ with thorns W^0 . Note that each $r \in \mathfrak{R}^0$ is vacuously 2-coherent, but not necessarily 2-saturated. A 1-cactus is defined symmetrically.

induction step Let $((W, R_1, R_2), w_0, \mathfrak{R})$ be a 2-cactus for Φ with thorns V and $\mathfrak{Q}^v$, for $v \in V$, a collection of 2-quasimodels $\mathfrak{Q}^v = ((W^v, R_2^v), v, \mathfrak{R}^v)$ for Φ such that, for each $v \in V$, we have (1) and

$$\mathfrak{R}(v) = \mathfrak{R}^v(v).$$

Then $((W', R_1, R_2 \cup \bigcup_{v \in V} R_2^v), w_0, \mathfrak{R}')$ is a 1-cactus for Φ with thorns $W' \setminus W$, where $W' = W \cup \bigcup_{v \in V} W^v$,

$$\mathfrak{R}' = \bigcup_{r \in \mathfrak{R}} \{r \cup \bigcup_{v \in \text{dom } r \cap V} r^v \mid r^v \in (\mathfrak{R}^v)_v \text{ with } r(v) = r^v(v), \text{ for } v \in \text{dom } r \cap V\} \cup \bigcup_{v \in V} (\mathfrak{R}^v \setminus (\mathfrak{R}^v)_v),$$

and $f \cup \bigcup_{v \in V} f^v$ denotes the partial function that coincides with f on $\text{dom } f$ and with f^v on $\text{dom } f^v$, for each $v \in V$; clearly, the Φ -runs in $\mathfrak{R}'$ are well-defined. We say that the $\mathfrak{Q}^v$ are *grafted* onto the 2-cactus. It can be seen that each $r' \in \mathfrak{R}'$ is 1- and 2-coherent as well as 2-saturated on W' , and is 1-saturated on W but not necessarily on $W' \setminus W$. Given a 1-cactus for Φ and a collection of 1-quasimodels, a 2-cactus for Φ is defined symmetrically.

A *limit i-cactus* for Φ is obtained as the limit of the sequence of cacti starting from a *i-quasimodel*.

We construct a limit 1-cactus for Φ by using the following procedure. We first observe that, by (G3.2) and global Kripke completeness of L_1 , we have $\varphi^1 \in t^1$ and so $\varphi \in t$ and thus $\varphi^2 \in t^2$, for every $t \in \mathbf{q}$ and $\mathbf{q} \in \mathbf{Q}$. We then begin, by (G3.1) and global Kripke completeness of L_1 , with an *xd-model* $\mathfrak{M}^0$ based on $(W^0, R_1^0) \in \mathcal{C}_1$ such that $\mathfrak{M}^0, w \models \varphi^1$ and $\mathfrak{M}^0, w \models \hat{\mathbf{Q}}^1$, for all $w \in W^0$, but $\mathfrak{M}^0, w_0 \not\models \psi^1$, for some $w_0 \in W^0$. With each $w \in W^0$ and $e \in \Delta_w^0$, we associate a uniquely determined Φ -type $t_{w,e}$ such that

$$\mathfrak{M}^0, w \models t_{w,e}^1[e].$$

Let $\mathfrak{R}^0$ be the set of functions r_e , for $e \in \bigcup_{w \in W^0} \Delta_w^0$, that map each $w \in W^0$ with $e \in \Delta_w^0$ to the type $t_{w,e}$. It follows that each r_e is a 1-coherent and 1-saturated Φ -run. The 1-quasimodel $((W^0, R_1^0), w_0, \mathfrak{R}^0)$ for Φ gives rise to our initial 2-cactus for Φ with thorns W^0 . Observe that, for every $\chi \in \Phi$ and $i = 1, 2$, we have

$$\mathfrak{M}^0, w \models \chi^i[e] \quad \text{iff} \quad \chi^i \in t_{w,e}^i, \quad \text{for all } w \in W^0 \text{ and } e \in \Delta_w^0.$$

Let $\mathfrak{C} = ((W, R_1, R_2), w_0, \mathfrak{R})$ be a 2-cactus for Φ with thorns $V \subseteq W$. Take any $v \in V$ and consider the Φ -quasistate $\mathfrak{R}(v) = \mathbf{q}^v \in \mathbf{Q}$ selected for v . By (G3.3) and global Kripke completeness of L_2 , there is an

xd -model $\mathfrak{M}^v$ based on $(W^v, R_2^v) \in \mathcal{C}_2$ such that $\mathfrak{M}^v, v \models (\hat{q}^v)^2$ and $\mathfrak{M}^v, w \models \hat{Q}^2$, for all $w \in W^v$. Similarly to the previous step, with each $w \in W^v$ and $e \in \Delta_w^v$, we associate a uniquely determined Φ -type $t_{w,e}^v$ such that

$$\mathfrak{M}^v, w \models (t_{w,e}^v)^2[e].$$

Let $\mathfrak{R}^v$ be the set of functions r_e^v , for $e \in \bigcup_{w \in W^v} \Delta_w^v$, that map each $w \in W^v$ with $e \in \Delta_w^v$ to the type $t_{w,e}^v$. We have $\mathfrak{R}^v(v) = \mathbf{q}^v = \mathfrak{R}(v)$. As before, $\mathfrak{Q}^v = ((W^v, R_2^v), v, \mathfrak{R}^v)$, for each thorn $v \in V$, is a 2-quasimodel for Φ . We can assume that the W^v satisfy (1), and so we extend the 2-cactus $\mathfrak{C}$ to a 1-cactus $\mathfrak{C}'$ for Φ by grafting 2-quasimodels $\mathfrak{Q}^v$ onto $\mathfrak{C}$.

By repeating this procedure, we obtain a limit 1-cactus $\mathfrak{C} = ((W, R_1, R_2), w_0, \mathfrak{R})$ for Φ . Since the $\mathfrak{M}^v$ realise the respective quasimodels, for every $\chi \in \Phi$, $i = 1, 2$ and $v \in W$, we have

$$\mathfrak{M}^v, w \models \chi^i[e] \quad \text{iff} \quad \chi^i \in (r_e(w))^i, \quad \text{for all } w \in W^v \text{ and } e \in \Delta_w^v.$$

We now construct a Kripke model $\mathfrak{M}$ from the limit 1-cactus $\mathfrak{C}$ for Φ . The frame is taken from $\mathfrak{C}$ (by assumption, both $\mathcal{C}_1$ and $\mathcal{C}_2$ are closed under disjoint unions), the domain $\Delta_w = \mathfrak{R}_w$, for each $w \in W$, and $\mathfrak{M}, w \models P[r]$ iff $P(x) \in r(w)$, for every unary predicate letter P , $w \in W$ and $r \in \mathfrak{R}_w$. By induction on the structure of $\chi \in \Phi$, we show that, for each $w \in W$ and $r \in \mathfrak{R}_w$,

$$\mathfrak{M}, w \models \chi[r] \quad \text{iff} \quad \chi^1 \in (r(w))^1 \quad \text{iff} \quad \chi^2 \in (r(w))^2.$$

The basis of induction is by definition. The cases of the Boolean connectives and quantifiers follow from the definition of quasistates. The cases of the modalities follow from the fact that the Φ -runs in $\mathfrak{R}$ are i -coherent and i -saturated. Thus, $\mathfrak{M}, w \models \varphi$, for all $w \in W$, but $\mathfrak{M}, w_0 \not\models \psi$. $\square$

The claims in Theorem 3.1 for cd -models are obtained by a straightforward modification of the construction in Lemma 3.5 (the runs are required to be total functions on the sets of worlds).

We now turn to the case of local consequence, which is technically more elaborate. The construction is also inspired by the cactus (quasi)models, but instead of starting from quasimodels and grafting more quasimodels to the cactus to obtain the required (quasi)model in the limit, the proof for the local consequence proceeds in the opposite direction: it starts from the ‘leaves’ and grafts smaller cacti onto a quasimodel. In fact, this proof requires only a finite number of steps, which depends on the so-called alternation depth (which is formally defined below, but, intuitively, reflects the number of switches between the modalities in the formula), while the ‘relevant’ part of types in quasimodels gets smaller and smaller as we move away from the world where the given formula is to be satisfied.

For an $\mathcal{L}$ -formula φ , consider the following set of subformulas:

$$\Theta^1(\varphi) = \{P(x) \mid P(x) \in \text{sub } \varphi\} \cup \{\chi \in \text{sub } \Box_2 \psi \mid \Box_2 \psi \in \text{sub } \varphi\}.$$

It can be thought of as removing the outermost ‘layer’ of Booleans, quantifiers and $\Box_1$ -operators. The set $\Theta^2(\varphi)$ is defined symmetrically. For a set Φ of $\mathcal{L}$ -formulas, let $\Theta^i(\Phi) = \bigcup_{\varphi \in \Phi} \Theta^i(\varphi)$.

We now define the notion of *alternation depth* of a formula using a converse operation to Θ^i . More precisely, for an $\mathcal{L}$ -formula ψ with surrogates for $\Box_2$ -subformulas, we denote

$$\psi \uparrow^1 = \psi[(\Box_2 \chi)^1 / (\Box_2 \chi)^2 \mid (\Box_2 \chi)^1 \in \text{sub } \psi].$$

In other words, operation $\cdot \uparrow^1$ restores the ‘meaning’ of the surrogates by re-introducing their $\Box_2$ -modalities, but the replacement subformulas are built from surrogates for their own $\Box_1$ -subformulas instead

(and so contain no $\Box_1$ -modalities themselves); $\psi \uparrow^2$ is defined symmetrically. It should be clear that, if these operations are applied repeatedly, alternating between $\cdot \uparrow^1$ and $\cdot \uparrow^2$, then we arrive at the original formula without surrogates: more precisely, for any $\mathcal{L}$ -formula φ , we have $\varphi = (\varphi^1) \uparrow^1 \uparrow^2 \dots \uparrow^i$, for some $i \in \{1, 2\}$. Denote by $\text{adp}^1(\varphi)$ the minimum length of such a sequence; symmetrically, $\text{adp}^2(\varphi)$ is the smallest number of $\cdot \uparrow^2$ and $\cdot \uparrow^1$ operations such that $\varphi = (\varphi^2) \uparrow^2 \uparrow^1 \dots \uparrow^i$. For example, $\text{adp}^1(\Box_1 \Box_2 p) = 1$ and $\text{adp}^1(\Box_2 \Box_2 \Box_1 r) = 2$; thus, $\text{adp}^1(\Box_1 \Box_2 p \wedge \Box_2 \Box_2 \Box_1 r) = 2$. It follows from the definitions that

$$|\text{adp}^1(\varphi) - \text{adp}^2(\varphi)| \leq 1,$$

depending on the leads of the longest alternation sequences in φ . Denote $\text{adp}^i(\Phi) = \text{adp}^i(\bigwedge_{\varphi \in \Phi} \varphi)$. Let $\text{adp}(\varphi) = \min(\text{adp}^1(\varphi), \text{adp}^2(\varphi))$ and $\text{adp}(\Phi) = \text{adp}(\bigwedge_{\varphi \in \Phi} \varphi)$.

Observe that, for any set Φ of $\mathcal{L}$ -formulas and $i = 1, 2$ with $\text{adp}(\Phi) = \text{adp}^i(\Phi)$, we have

$$\text{adp}(\Theta^i(\Phi)) = \max\{0, \text{adp}(\Phi) - 1\} \quad \text{and} \quad \text{adp}(\Theta^i(\Phi)) = \text{adp}^{3-i}(\Theta^i(\Phi)). \quad (\dagger)$$

In addition to adp , we define the *modal 1-depth* $md^1(\varphi)$ of φ as the maximum number of nested $\Box_1$ (ignoring $\Box_2$); the *modal 2-depth* $md^2(\varphi)$ is defined symmetrically. We also set $md^i(\Phi) = md^i(\bigwedge_{\varphi \in \Phi} \varphi)$. The modal i -depth will be used to describe the decreasing number of ‘relevant’ formulas as we move away from the world witnessing the formula, while the alternation depth will be used to count the number of requires R_1 - and R_2 -components along the paths.

Finally, for an $\mathcal{L}$ -formula φ , consider

$$\mathcal{Q}_i(\varphi) = \{q \text{ a } \Theta^i(\varphi)\text{-quasistate such that } \not\models_{L_1 \otimes L_2} \neg \hat{q}\},$$

and, as before, let $\hat{\mathcal{Q}}_i(\varphi)$ denote the disjunction of all $\hat{q}$, for $q \in \mathcal{Q}_i(\varphi)$.

With these definitions at hand, we can formulate our criterion for deciding local consequence, as claimed in Theorem 3.1.

Lemma 3.8. *Let $\mathcal{C}_1, \mathcal{C}_2$ be non-empty classes of frames closed under disjoint unions and $L_i = \text{Log}_{\text{xd}} \mathcal{C}_i$, for both $i = 1, 2$. For all $\mathcal{L}$ -formulas φ and $i = 1, 2$ with $\text{adp}(\varphi) = \text{adp}^i(\varphi)$, the following are equivalent:*

- (L1) $\not\models_{L_1 \otimes L_2} \varphi$;
- (L2) *there is an xd-model $\mathfrak{M}$ based on a $\mathcal{C}_1 \otimes \mathcal{C}_2$ -frame such that $\mathfrak{M} \not\models \varphi$;*
- (L3) $\not\models_{L_i} \Box_i^{\leq md^i(\varphi)} (\hat{\mathcal{Q}}_i(\varphi))^i \rightarrow \varphi^i$.

Completeness of $L_1 \otimes L_2$ for $\mathcal{C}_1 \otimes \mathcal{C}_2$ claimed in Theorem 3.1 then follows immediately. For decidability, observe that, while the definition of $\mathcal{Q}_i(\varphi)$ refers to $\vdash_{L_1 \otimes L_2}$ and thus may appear circular, by property $(\dagger)$, the $\Theta^i(\varphi)$ -formulas have smaller alternation depth, and so we can apply the criterion in Lemma 3.8 recursively to the simpler formulas to decide whether the respective q should be included in the set $\mathcal{Q}_i(\varphi)$. This recursion terminates as we eventually reach formulas of alternation depth 0, which, by definition, belong to either $\mathcal{L}_1$ or $\mathcal{L}_2$, where, by assumption, the consequence relation is decidable.

Definition 3.9. Let Φ be a set of $\mathcal{L}$ -formulas closed under subformulas, For $i = 1, 2$ and $h \leq md^i(\Phi)$, denote $\Gamma_h^i(\Phi) = \{\psi \in \Phi \mid md^i(\psi) \leq h\}$. Since $\Gamma_h^i(\Phi) = \emptyset$ for $h < 0$, there is only one such $\Gamma_h^i(\Phi)$ -type, $\emptyset$. For a type t , let $t|_\Phi$ be the restriction of t to Φ , that is, $t \cap \{\psi, \neg \psi \mid \psi \in \Phi\}$, and denote $q|_\Phi = \{t|_\Phi \mid t \in q\}$.

Let $(W, R_i) \in \mathcal{C}_i$. For $w, w' \in W$, let $\text{dist}_{R_i}(w, w')$ be the length of the shortest R_i -path from w to w' . A *tapered Φ -run through* $((W, R_i), w_0)$ is a function r from an R_i -closed subset of W , denoted by $\text{dom } r$, such that each $r(w)$ is a $\Gamma_{h(w)}^i(\Phi)$ -type, where $h(w) = md^i(\Phi) - \text{dist}_{R_i}(w_0, w)$. For $w \in \text{dom } r$, run r is

- *i-coherent at w* if $\Box_i \psi \in r(w)$ implies $\psi \in r(w')$, for all $w' \in R_i(w)$;

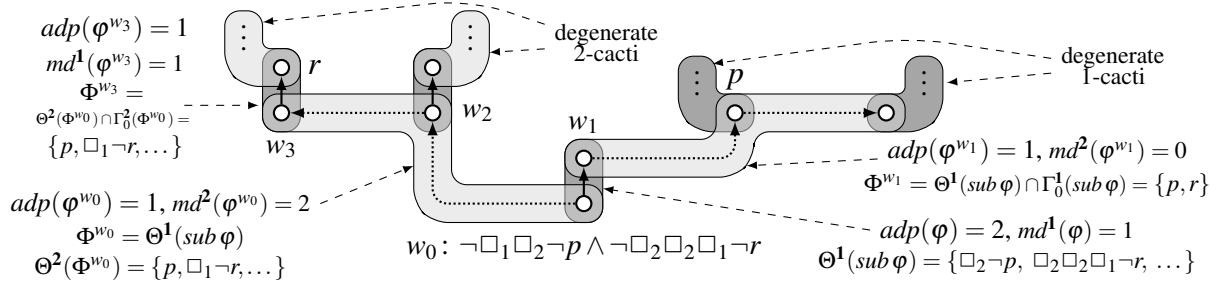


Figure 2: A fully grafted tapered 1-cactus satisfying $\phi = \Diamond_1 \Diamond_2 p \wedge \Diamond_2 \Diamond_2 \Diamond_1 r$: solid arrows with dark background for R_1 -components and dotted arrows with light background for R_2 -components.

- *i-saturated at w* if $\neg \Box_i \psi \notin r(w)$ whenever $\psi \in r(w')$, for all $w' \in R_i(w)$.³

We say that r is *i-coherent on* $W' \subseteq W$ if r is *i-coherent* at each $w \in \text{dom } r \cap W'$ and *i-coherent* if r is *i-coherent* on W ; similarly for r being *i-saturated*.

A *tapered i-quasimodel* Ω for Φ is a tuple of the form $((W, R_i), w_0, \mathfrak{R})$, where $(W, R_i) \in \mathcal{C}_i$, $w_0 \in W$, and $\mathfrak{R}$ is a set of tapered Φ -runs through $((W, R_i), w_0)$ such that each $r \in \mathfrak{R}$ is *i-coherent* and *i-saturated* and each $\mathfrak{R}(w)$ is a Φ -quasistate, for $w \in W$, with $\mathfrak{R}_w = \{r \in \mathfrak{R} \mid w \in \text{dom } r\}$ and $\mathfrak{R}(w) = \{r(w) \mid r \in \mathfrak{R}_w\}$.

Definition 3.10. We inductively define a *tapered i-cactus* $\mathfrak{C}$ for Φ as a tuple of the form $((W, R_1, R_2), w_0, \mathfrak{R})$, where $(W, R_i) \in \mathcal{C}_i$ and $(W \setminus \{w_0\}, R_{3-i}) \in \mathcal{C}_{3-i}$ with a distinguished world $w_0 \in W$, which is not in the domain or range of R_{3-i} , and $\mathfrak{R}$ is a set of tapered Φ -runs that are *i-* and $(3-i)$ -coherent as well as *i-saturated* on W but $(3-i)$ -saturated only on $W \setminus \{w_0\}$; see Fig. 2, which shows the frame structure.

basis: $\Phi = \emptyset$. A tapered ‘degenerate’ 1-cactus $((W, R_1, R_2), w_0, \mathfrak{R})$ for $\emptyset$ is constructed by first taking sufficiently large disjoint unions (W, R_1) and (W', R'_2) of copies of $\mathfrak{F}_1 \in \mathcal{C}_1$ and $\mathfrak{F}_2 \in \mathcal{C}_2$, respectively, such that $|W| - 1 = |W'|$. We then take any $w_0 \in W$ and obtain the required (W, R_1, R_2) by applying to R'_2 the bijection between $W \setminus \{w_0\}$ and W' . Finally, $\mathfrak{R}$ consists of a single tapered $\emptyset$ -run r with $r(w) = \emptyset$, for all $w \in W$, which is trivially 1- and 2-coherent and 1- and 2-saturated.

induction step: $\Phi \neq \emptyset$. Let $\Omega = ((W^0, R_1^0), w_0, \mathfrak{R}^0)$ be a tapered 1-quasimodel for Φ . For each $v \in W^0$, denote

$$\Phi^v = \Theta^1(\Phi) \cap \Gamma_{h(v)}^1(\Phi), \quad \text{where } h(v) = md^1(\Phi) - dist_{R_1^0}(w_0, v).$$

For each $v \in W^0 \setminus \{w_0\}$, take a tapered 2-cactus $\mathfrak{C}^v = ((W^v, R_1^v, R_2^v), v, \mathfrak{R}^v)$ for Φ^v that is *compatible* with Ω :

$$W^0 \cap W^v = \{v\} \quad \text{and} \quad \mathfrak{R}^0(v)|_{\Theta^1(\Phi)} = \mathfrak{R}^v(v).$$

We also assume that the W^v are disjoint. Then $((W, R_1, R_2), w_0, \mathfrak{R})$ is a *tapered 1-cactus* $\mathfrak{C}$ for Φ if it is obtained by grafting the $\mathfrak{C}^v$, for $v \in W^0 \setminus \{w_0\}$, onto Ω . More precisely, we say that $((W^0, R_1^0, \emptyset), w_0, \mathfrak{R}^0)$ is an initial *stock* and W^0 its *thorns*. Then, given a stock $((W, R_1, R_2), w_0, \mathfrak{R})$ and a thorn $v \in W$, the *grafting of* $\mathfrak{C}^v$ *onto* v is a stock $((W \cup W^v, R_1 \cup R_1^v, R_2 \cup R_2^v), w_0, \mathfrak{R}')$, where

$$\mathfrak{R}' = \mathfrak{R} \setminus \mathfrak{R}_v \cup \{r \cup r^v \mid r \in \mathfrak{R}_v \text{ and } r^v \in (\mathfrak{R}^v)_v \text{ with } r(v)|_{\Theta^1(\Phi)} = r^v(v)\} \cup \mathfrak{R}^v \setminus (\mathfrak{R}^v)_v.$$

Observe that, as v in $\mathfrak{C}^v$ is not R_1 -connected, the R_1 -components in Ω and $\mathfrak{C}^v$ are disjoint, and so the frame in the resulting stock belongs to $\mathcal{C}_1 \otimes \mathcal{C}_2$. The resulting runs are 1- and 2-coherent and 1-saturated on $W \cup W^v$. The runs are 2-saturated on $W' \cup W^v$ if they are 2-saturated on $W' \subseteq W$.

³We use $\neg \Box_i \psi \notin r(w)$ as a convenient replacement for either $\Box_i \psi \notin \Gamma_{h(w)}^i(\Phi)$ or $\Box_i \psi \in r(w)$.

A fully grafted tapered i -cactus $((W, R_1, R_2), \mathfrak{R})$ for Φ results from grafting a compatible tapered $(3-i)$ -cactus $\mathfrak{C}^{w_0}$ for $\Phi^{w_0} = \Theta^i(\Phi)$ onto the world w_0 of a tapered i -cactus $((W^0, R_1^0, R_2^0), w_0, \mathfrak{R}^0)$ for Φ . Runs in a fully grafted tapered cactus are i -coherent and i -saturated for both $i = 1, 2$ on the whole W .

Claim 3.11. For all $\mathcal{L}$ -formulas φ and $i = 1, 2$ with $\text{adp}(\varphi) = \text{adp}^i(\varphi)$, each of **(L1)**–**(L3)** holds iff **(L4)** there is a (fully grafted) tapered i -cactus $((W, R_1, R_2), \mathfrak{R})$ for $\text{sub } \varphi$ with $\varphi \notin r(w_0)$, for some $r \in \mathfrak{R}$.

Proof. We proceed by induction on alternation depth. The basis, $\text{adp}(\varphi) = 0$, is trivial as the formula belongs to $\mathcal{L}_1$ or $\mathcal{L}_2$ (or both, if it has no modalities). For the inductive step, consider an $\mathcal{L}$ -formula φ and suppose that $\text{adp}(\varphi) = k > 0$. Assume that $\text{adp}^1(\varphi) = k$ (the other case is symmetrical) and that the claim has been shown for all formulas ψ with $\text{adp}^2(\psi) < k$, which, by $(\dagger)$, includes all $\psi \in \Theta^1(\varphi)$.

While **(L2)** $\Rightarrow$ **(L1)** follows from soundness of $\vdash_{L_1 \otimes L_2}$, **(L1)** $\Rightarrow$ **(L3)** follows from closure under substitutions of $\vdash_{L_i}$ (cf. **(G1)** $\Rightarrow$ **(G3)** in the proof of Lemma 3.5).

(L3) $\Rightarrow$ **(L4)** By Kripke completeness of L_1 , there is an xd -model $\mathfrak{M}^0$ based on $(W^0, R_1^0) \in \mathcal{C}_1$ such that $\mathfrak{M}^0, w_0 \models \Box_1^{<\text{adp}^1(\varphi)}(\hat{\mathcal{Q}}_1(\varphi))^1$ and $\mathfrak{M}^0, w_0 \not\models \varphi^1$ for some $w_0 \in W^0$. Let $\Phi = \text{sub } \varphi$. For each $v \in W^0$, let $h(v) = \text{md}^1(\Phi) - \text{dist}_{R_1^0}(w_0, v)$. Construct a set $\mathfrak{R}^0$ of tapered Φ -runs through $((W^0, R_1^0), w_0)$: for each $e \in \bigcup_{v \in W^0} \Delta_v^0$, consider r_e that maps each $v \in W^0$ with $e \in \Delta_v^0$ to the $\Gamma_{h(v)}^1(\Phi)$ -type $t_{v,e}$ with $\mathfrak{M}, v \models t_{v,e}^1[e]$. Thus, $\mathfrak{Q} = ((W^0, R_1^0), w_0, \mathfrak{R}^0)$ is a tapered 1-quasimodel for Φ with $\varphi \notin r_e(w_0)$, for some $e \in \Delta_{w_0}^0$.

For each $v \in W^0$ with $h(v) \geq 0$, take the $\Theta^1(\Phi)$ -quasistate $\mathbf{q}^v \in \mathcal{Q}_1(\varphi)$ with $\mathfrak{M}^0, v \models (\hat{\mathbf{q}}^v)^1$. Denote $\Phi^v = \Theta^1(\Phi) \cap \Gamma_{h(v)}^1(\Phi)$ and consider $\mathbf{q}^v|_{\Gamma_{h(v)}^1(\Phi)}$, which is a Φ^v -quasistate with $\mathbf{q}^v|_{\Gamma_{h(v)}^1(\Phi)} = \mathfrak{R}^0(v)|_{\Theta^1(\Phi)}$. Let φ^v be its realisability sentence. As $\text{adp}^2(\varphi^v) < k$, we apply IH to φ^v and obtain a tapered 2-cactus $\mathfrak{C}^v = ((W^v, R_1^v, R_2^v), v, \mathfrak{R}^v)$ for Φ^v such that $\mathbf{q}^v|_{\Gamma_{h(v)}^1(\Phi)} = \mathfrak{R}^v(v)$. For each $v \in W^0$ with $h(v) < 0$, take a tapered 2-cactus $\mathfrak{C}^v$ for $\emptyset$. So, for each $v \in W^0 \setminus \{w_0\}$, we graft $\mathfrak{C}^v$ onto v in $\mathfrak{Q}$ to obtain a tapered 1-cactus $\mathfrak{C}$. A fully grafted tapered 1-cactus can be obtained by grafting, in addition, $\mathfrak{C}^{w_0}$ onto w_0 in $\mathfrak{C}$.

(L4) $\Rightarrow$ **(L2)** Let $\mathfrak{C} = ((W, R_1, R_2), \mathfrak{R})$ be a fully grafted tapered i -cactus for $\text{sub } \varphi$ with $\varphi \notin r(w_0)$, for some $r \in \mathfrak{R}$. We construct $\mathfrak{M}$ based on a $\mathcal{C}_1 \otimes \mathcal{C}_2$ -frame (W, R_1, R_2) and domains $\Delta_w = \mathfrak{R}_w$, for $w \in W$: we set $\mathfrak{M}, w \models P[r]$ iff $P(x) \in r(w)$, for each unary predicate letter P , $w \in W$ and $r \in \mathfrak{R}_w$. By induction on the structure of formulas, we show that, for each $w \in W$, $r \in \mathfrak{R}_w$, $\chi \in \{\psi, \neg\psi \mid \psi \in r(w)\}$, and $i = 1, 2$,

$$\mathfrak{M}, w \models \chi[r] \quad \text{iff} \quad \chi^i \in (r(w))^i.$$

The basis of induction is by definition. The cases of the Boolean connectives and quantifiers are immediate from the definition of types and quasistates. The cases of the modalities follow from the fact that the runs in $\mathfrak{C}$ are i -coherent and i -saturated. Thus, $\mathfrak{M}, w_0 \not\models \varphi$. $\square$

The claims on local consequence in Theorem 3.1 for cd -models are obtained by a straightforward modification of the construction in Lemma 3.8 (the tapered runs are required to be total functions on the sets of worlds). Finally, the transfer of the d -fmp follows from the proof of Claim 3.11 (the fully grafted cactus is finite).

4 Fusions of One-Variable Logics with Equality

We show that once equality is allowed in one-variable modal logics, the transfer results established in Section 3 no longer hold. Remarkably, the counterexamples we introduce next arise quite naturally. The propositional modal logic **Diff** of the ‘elsewhere’ operator was introduced by Von Wright [34] and is determined by the class $\mathcal{D}$ of all disjoint unions of frames (W, R) with $R = \{(w, v) \in W \times W \mid w \neq v\}$; see

also [27]. We are interested here in the one-variable modal logics $\text{Log}_{cd}^{\bar{}} \mathcal{D}$ and $\text{Log}_{cd}^{\bar{}} \mathcal{D}_{\text{fin}}$ with $\mathcal{D}_{\text{fin}}$ the class of disjoint unions of *finite* frames in $\mathcal{D}$. Note that the propositional logic **Diff** has the fmp, and so is complete for both $\mathcal{D}$ and $\mathcal{D}_{\text{fin}}$. On the other hand, while one can easily prove the fmp for $\text{Log}_{cd}^{\bar{}} \mathcal{D}_{\text{fin}}$, the logic $\text{Log}_{cd}^{\bar{}} \mathcal{D}$ does not enjoy the fmp, and so $\text{Log}_{cd}^{\bar{}} \mathcal{D} \subsetneq \text{Log}_{cd}^{\bar{}} \mathcal{D}_{\text{fin}}$ [18]. It has been observed [17, 18] that both logics can be reduced in polytime to the two-variable fragment of first-order logic with counting on arbitrary and finite domains, respectively, and so they are decidable in CONEXPTIME.

Theorem 4.1. *Both $\text{Log}_{cd}^{\bar{}} \mathcal{D}$ and $\text{Log}_{cd}^{\bar{}} \mathcal{D}_{\text{fin}}$ are decidable but $\text{Log}_{cd}^{\bar{}}(\mathcal{D} \otimes \mathcal{D}_{\text{fin}})$ is undecidable.*

Proof. The proof is by reduction of the problem of solving Diophantine equations. For arbitrary polynomials g, h with coefficients from $\mathbb{N} \setminus \{0\}$, it is undecidable whether $g = h$ has a solution in $\mathbb{N} \setminus \{0\}$ [7]. Each polynomial equation can be rewritten into a set of elementary equations $y = n$, $y = z_1 + z_2$, and $y = z_1 \times z_2$, with variables y, z_1, z_2 and $n \in \mathbb{N} \setminus \{0\}$. So we assume that a set $E = \{e_1, \dots, e_n\}$ of such equations is given. We construct a formula φ_E which is satisfiable iff E has a solution in $\mathbb{N} \setminus \{0\}$. Let the modal operators $\Box$ and $\Box_C$ be interpreted over frames with accessibility relations R in $\mathcal{D}$ and R_C in $\mathcal{D}_{\text{fin}}$, respectively. We set $\Box^+ \varphi = \varphi \wedge \Box \varphi$, and likewise for $\Box_C$. Formula φ_E will hardly need any alternation between modalities, the maximal alternation being of the form $\Box \Box_C \chi$. In fact, the only contribution of $\mathcal{D}_{\text{fin}}$ is to supply the required *finite* cardinality of the connected components of its frames.

When discussing the construction of φ_E , we assume a world w in the set W of worlds reachable along R . To W attached are the models for $\Box_C$ with the R_C -connected component of w denoted by W_w . For a unary predicate letter P and $w \in W$, we denote by $|P|_w$ the cardinality of the extension of P at w . In the sequel, for a closed formula I and a unary predicate P , we use formula $\text{card}(I, P)$ defined as follows:

$$\Box^+ [I \rightarrow \Box_C^+ \forall x ((x = c) \rightarrow \Box_C (x \neq c)) \wedge \forall x (P(x) \leftrightarrow \Diamond_C^+ (x = c))],$$

where c is a fresh constant. Formula $\text{card}(I, P)$ ensures that, at any world w where I holds, the number of worlds in the R_C -connected component of w coincides with the number of domain elements in the extension of P at w , that is, $|W_w| = |P|_w$; see Fig. 3 a), where the nodes representing the same domain element in different worlds are shown connected by the R - and R_C -relations.

For each variable y used on the left- or right-hand side of equations in E , we take unary predicate letters P_y and C_y . Intuitively, $|P_y|_w$ encodes the value of y , while C_y is used to ensure that the value of y is finite. To this end, φ_E contains the conjunct

$$\Diamond^+ (I_y \wedge \Box \neg I_y), \quad \text{where } I_y = \exists x C_y(x), \quad (2)$$

to mark a single world $w \in W$ with I_y . We add the conjunct

$$\text{card}(I_y, P_y) \quad (3)$$

to φ_E to ensure that $|P_y|_w = |W_w|$. Note that $|P_y|_w$ is finite. The following conjuncts guarantee that each P_y is non-empty (only solutions in $\mathbb{N} \setminus \{0\}$ are admitted) and constant on W :

$$\exists x P_y(x) \quad \text{and} \quad \Box^+ \forall x (P_y(x) \rightarrow \Box P_y(x)). \quad (4)$$

- If $e \in E$ is of the form $y = n$, then we take n constants $c_1^e, \dots, c_n^e$ interpreted differently and make the set of their interpretations the extension of P_y :

$$\forall x (P_y(x) \leftrightarrow (x = c_1^e) \vee \dots \vee (x = c_n^e)) \wedge \bigwedge_{i \neq j} (c_i^e \neq c_j^e). \quad (5)$$

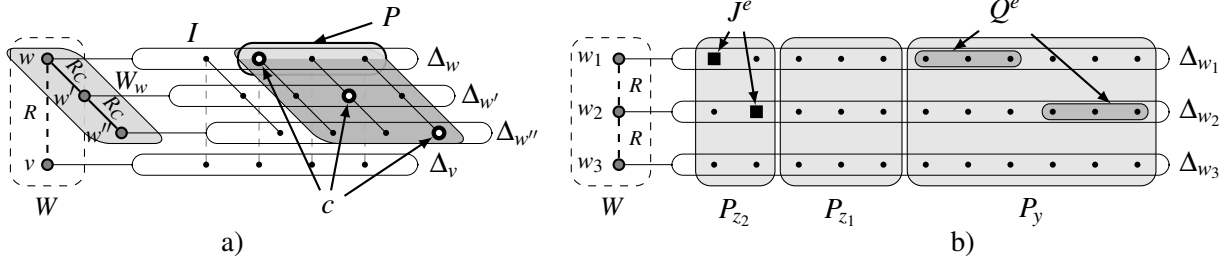


Figure 3: a) Formula $\text{card}(I, P)$. b) Equation e of the form $y = z_1 \times z_2$ with $z_1 = 3$ and $z_2 = 2$.

- If $e \in E$ is of the form $y = z_1 + z_2$, we introduce unary predicate letters Q_1^e and Q_2^e and add formulas

$$\text{card}(I_{z_i}, Q_i^e) \quad \text{for } i = 1, 2, \quad (6)$$

which ensure $|P_{z_i}|_{w_i} = |Q_i^e|_{w_i}$ at the world w_i marked with I_{z_i} . Next, we state that Q_1^e and Q_2^e are disjoint at all worlds $w \in W$ and empty at worlds $w \in W$ where I_{z_i} does not hold:

$$\Box^+ \forall x (Q_1^e(x) \rightarrow \Box^+ \neg Q_2^e(x)), \quad \Box^+ (\exists x Q_1^e(x) \rightarrow I_{z_1}) \quad \text{and} \quad \Box^+ (\exists x Q_2^e(x) \rightarrow I_{z_2}). \quad (7)$$

Finally, we add the conjunct that ensures that P_y is the union of Q_1^e and Q_2^e :

$$\forall x (P_y(x) \leftrightarrow \Diamond^+ Q_1^e(x) \vee \Diamond^+ Q_2^e(x)), \quad (8)$$

which implies that equation e holds in the sense that $|P_y|_w = |P_{z_1}|_w + |P_{z_2}|_w$, for all worlds $w \in W$.

- If $e \in E$ is of the form $y = z_1 \times z_2$, then we introduce a unary predicate letter C^e and a constant c^e and let $J^e(x) = C^e(x) \wedge (c^e = x)$. First, we ensure $|P_{z_2}|_w$ is equal to the number of worlds in W satisfying $\exists x J^e(x)$:

$$\forall x (P_{z_2}(x) \leftrightarrow \Diamond^+ J^e(x)) \quad \text{and} \quad \Box^+ \forall x (J^e(x) \rightarrow \Box \neg J^e(x)), \quad (9)$$

where, due to the second conjunct, $J^e(x)$ cannot hold on the same domain element more than once in W . Next, we introduce a unary predicate letter Q^e and add formulas

$$\text{card}(\exists x J^e(x), P_{z_1}) \quad \text{and} \quad \text{card}(\exists x J^e(x), Q^e) \quad (10)$$

to ensure that $|P_{z_1}|_w = |Q^e|_w$ for all worlds $w \in W$ with $\exists x J^e(x)$. Then we state that the Q^e are mutually disjoint at different worlds $w \in W$ and empty at worlds $w \in W$ where $\exists x J^e(x)$ does not hold:

$$\Box^+ \forall x (Q^e(x) \rightarrow \Box \neg Q^e(x)) \quad \text{and} \quad \Box^+ (\exists x Q^e(x) \rightarrow \exists x J^e(x)). \quad (11)$$

We have $|P_{z_1}|_w$ (encoded as $|Q^e|_w$) in exactly $|P_{z_2}|_w$ -many worlds and take the sum by adding the conjunct

$$\forall x (P_y(x) \leftrightarrow \Diamond^+ Q^e(x)), \quad (12)$$

which implies that equation e holds in the sense that $|P_y|_w = |P_{z_1}|_w \times |P_{z_2}|_w$, for all $w \in W$; see Fig. 3 b), where only worlds in W and their domains are shown, and $|W_{w_1}| = |W_{w_2}| = 2$.

So, φ_E contains (2)–(4), for each variable y in equations in E , and one of (5), (6)–(8) or (9)–(12), for each equation $e \in E$. It should be clear that φ_E is as required. $\square$

Theorem 4.1 does not depend on the constant domain semantics since $\mathcal{D}$ and $\mathcal{D}_{\text{fin}}$ do not feel the difference between expanding and constant domains. Note also that recursive axiomatisability is not preserved under fusions since we have shown that $\text{Log}_{\text{cd}}^-(\mathcal{D} \otimes \mathcal{D}_{\text{fin}})$ is not recursively enumerable.

For the global consequence, one can prove an even stronger non-transfer result.

Theorem 4.2. *Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be classes of Kripke frames that are closed under disjoint unions and contain non-trivial frames. Then the global consequence for $\text{Log}_{\text{cd}}^-(\mathcal{C}_1 \otimes \mathcal{C}_2)$ is undecidable.*

The proof is given in the full version of this paper available at [20]. It is by reduction of Minsky machines and very similar to the known undecidability proof for temporal first-order logic with equality [8].

We conclude this section with two remarks. First, note that neither $\mathcal{D}$ nor $\mathcal{D}_{\text{fin}}$ are propositionally modally definable classes of frames. In fact, as **Diff** is axiomatised by $p \rightarrow \Box \Diamond p$ and $\Diamond p \rightarrow \Box(p \vee \Diamond p)$, any frame (W, E) with an equivalence relation E validates **Diff** (in other words, **Diff** $\subseteq$ **S5**). The question arises whether our non-preservation result continues to hold when one restricts the components to propositionally modally definable frame classes. We conjecture that it does, but at present we only have a proof for expanding domain semantics and under the assumption that polymodal logics are permitted as components. To demonstrate this, let $\mathcal{C}_{\text{diff}}$ be the class of *all* frames validating **Diff** and $\mathcal{C}_{\text{ftime}}$ the class of all frames validating the bimodal logic **LTL_f** of all frames (W, S, R) with R a finite strict linear order and S its successor relation. **LTL_f** is axiomatised by taking the fusion of **GL3** and **Alt₁** and adding the standard induction axioms [15].

Theorem 4.3. *$\text{Log}_{\text{xd}}^-(\mathcal{C}_{\text{diff}})$ and $\text{Log}_{\text{xd}}^-(\mathcal{C}_{\text{ftime}})$ are decidable but $\text{Log}_{\text{xd}}^-(\mathcal{C}_{\text{diff}} \otimes \mathcal{C}_{\text{diff}} \otimes \mathcal{C}_{\text{ftime}})$ is undecidable.*

Proof. Decidability of $\text{Log}_{\text{xd}}^-(\mathcal{C}_{\text{diff}})$ is shown in [18]. For decidability (and Ackermann-hardness) of $\text{Log}_{\text{xd}}^-(\mathcal{C}_{\text{ftime}})$, see [19, 12]. Undecidability of $\text{Log}_{\text{xd}}^-(\mathcal{C}_{\text{diff}} \otimes \mathcal{C}_{\text{diff}} \otimes \mathcal{C}_{\text{ftime}})$ can be shown by adapting the reduction of Diophantine equations: first, observe that one can enforce that a frame in $\mathcal{C}_{\text{diff}}$ is actually in $\mathcal{D}$ using the formula

$$\Box^+ \forall x ((x = c) \rightarrow \Box(x \neq c)),$$

which can only be satisfied if all worlds in W are irreflexive. The fusion with $\mathcal{C}_{\text{ftime}}$ is used to enforce a *finite* first-order domain for worlds in W . This can be achieved as follows: for another fresh constant c' and $\Box_L$ the modal operator interpreted by the strict order in $\mathcal{C}_{\text{ftime}}$, if the sentence $\forall x \Box_L^+(x = c')$ is true at w_0 , then the domain at w_0 cannot be larger than the length of the finite strict order seen from w_0 . It is now straightforward to adapt the remaining steps of the undecidability proof. $\square$

We conjecture that using polymodal logics here is not essential. For instance, the standard reduction of polymodal logics to logics with one modality might be applicable [22].

Second, the undecidability proof can be adapted to resolve an open problem concerning monodic fragments of first-order modal logics [25, 1]. A key ingredient of the encoding of Diophantine equations is that we can enforce equal cardinality $|P|_w = |Q|_w$ of the extensions of predicates P and Q at world w . This can also be achieved directly using a bijective function in the extension of one-variable modal logic to, for instance, the monodic fragment of modal first-order logic over the two-variable fragment of **FO** with counting. Hence, it can be shown that this monodic fragment is undecidable over the class $\mathcal{D}_{\text{fin}}$. This shows that decidability does not always transfer from one-variable modal logic with equality to monodic fragments based on decidable fragments of first-order logic (recall that the two-variable fragment with counting is decidable [26]). Note that sufficient conditions for this kind of preservation from one-variable modal logics to monodic fragments are shown in [13].

Theorem 4.4. *$\text{Log}_{\text{cd}}^-(\mathcal{D}_{\text{fin}})$ is decidable, but the monodic fragment of the two-variable fragment with counting over $\mathcal{D}_{\text{fin}}$ is undecidable.*

5 Fusions of Propositional Modal Logics Sharing an S5 Modality

In this section, we generalise the fusion construction for one-variable first-order modal logics without equality to fusions of propositional modal logics sharing an **S5** modality. We begin by establishing a transfer theorem for logics admitting models that are homogeneous for the shared modality, and then apply it to (semi)commutators. In contrast to the transfer proof in Lemma 3.5, in the global case we do not construct cactus models; instead, we merge the respective models of the component logics in a single step to obtain models of the fusion. This approach works provided we can ensure that the extensions of the relevant types in models of the component logics have matching cardinalities. A closely related technique was first used in [33] to obtain transfer results via algebraic semantics rather than Kripke semantics, thereby avoiding any reliance on the Kripke completeness of the component logics.

Consider a propositional modal language with **S5** operator $\Box_E$ interpreted in Kripke models by the equivalence relation E . The E -equivalence class $[w]_E$ of $w \in W$ is defined as usual by setting $[w]_E = E(w)$. Let κ be an infinite cardinal. Then $\mathfrak{M}$ is called (E, κ) -homogeneous if, for every E -equivalence class U in $\mathfrak{F}$ and every modal formula φ , $|\{w \in U \mid \mathfrak{M}, w \models \varphi\}| \in \{0, \kappa\}$.

Definition 5.1. A propositional modal logic L with **S5** operator $\Box_E$ admits E -homogeneous models (for global consequence) if there exists an infinite cardinal κ such that, for all $\kappa' \geq \kappa$, whenever $\varphi \notin L$ (or, respectively, $\varphi \not\vdash_L^* \psi$), there exists an (E, κ') -homogeneous model $\mathfrak{M}$ based on an L -frame with $\mathfrak{M} \not\models \varphi$ (or, respectively, with $\mathfrak{M} \models \varphi$ and $\mathfrak{M} \not\models \psi$).

Note that modal logics that admit homogeneous models (for global consequence) are (globally) Kripke complete, by definition. To formulate our main results, assume for simplicity that L_i is a bi-modal logic in the propositional modal language $\mathcal{L}_{i,E}$ with modal operators $\Box_i$ and $\Box_E$, for $i = 1, 2$. Then $L_1 \otimes_E L_2$ denotes the smallest modal logic containing $L_1 \cup L_2$ in the language $\mathcal{L}_{1,2,E}$ with modal operators $\Box_1, \Box_2$, and $\Box_E$.

Theorem 5.2. Let L_1 and L_2 admit E -homogeneous models (for global consequence). Then

- $L_1 \otimes_E L_2$ is (globally) Kripke complete;
- $L_1 \otimes_E L_2$ is (globally) decidable if L_1 and L_2 are both (globally) decidable.

We give a detailed proof for the global consequence. The local case can be shown by a rather straightforward combination of the approaches to the local case in Section 3 and the global case in this section, see [20] for details. Assume L_1 and L_2 admit E -homogeneous models for global consequence.

Modulo replacing $\forall x$ with $\Box_E$, we use the same notation as in Section 3. For instance, if Φ is a finite set of $\mathcal{L}_{1,2,E}$ -formulas closed under subformulas, then a Φ -type is a maximal Boolean-consistent subset of $\{\psi, \neg\psi \mid \psi \in \Phi\}$, and a Φ -quasistate is a non-empty set of Φ -types such that

$$\Box_E \psi \in t \quad \text{iff} \quad \psi \in t', \text{ for all } t' \in \mathbf{q}, \quad \text{for each } \Box_E \psi \in \Phi \text{ and } t \in \mathbf{q}.$$

We set

$$\hat{\mathbf{q}} = \Box_E \bigvee_{t \in \mathbf{q}} t \wedge \bigwedge_{t \in \mathbf{q}} \Diamond_E t,$$

where, for simplicity, we use t to denote the conjunction of all formulas in Φ -type t . Again, $\hat{\mathbf{Q}}$ denotes $\bigvee \{\hat{\mathbf{q}} \mid \mathbf{q} \in \mathbf{Q}\}$ for a set $\mathbf{Q}$ of Φ -quasistates. We also define surrogates in the obvious way by introducing for each formula of the form $\Box_i \psi$, a fresh proposition letter $p_{\Box_i \psi}$ called the *surrogate* of $\Box_i \psi$. To establish the claimed results in the global case, it suffices to show the following lemma.

Lemma 5.3. Let φ, ψ be $\mathcal{L}_{1,2,E}$ -formulas. Denote $\Phi = \text{sub } \varphi \cup \text{sub } \psi$. Then the following are equivalent:

(G_E1) $\varphi \not\models_{L_1 \otimes_E L_2}^* \psi$;

(G_E2) *there is a model $\mathfrak{M}$ based on an $L_1 \otimes_E L_2$ -frame such that $\mathfrak{M} \models \varphi$ but $\mathfrak{M} \not\models \psi$;*

(G_E3) *there is a set $\mathcal{Q}$ of Φ -quasistates such that*

(G_E3.1) $\varphi^1 \wedge \hat{\mathcal{Q}}^1 \not\models_{L_1}^* \psi^1$,

(G_E3.2) $\varphi^1 \wedge \hat{\mathcal{Q}}^1 \not\models_{L_1}^* \neg \hat{q}_i^1$, for each $q_i \in \mathcal{Q}$,

(G_E3.3) $\hat{\mathcal{Q}}^2 \not\models_{L_2}^* \neg \hat{q}_i^2$, for each $q_i \in \mathcal{Q}$.

Proof. (G_E2) $\Rightarrow$ (G_E1) and (G_E1) $\Rightarrow$ (G_E3) are shown similar to the proof of Lemma 3.5.

(G_E3) $\Rightarrow$ (G_E2) Let $\mathcal{Q}$ be a set of Φ -quasistates such that (G_E3.1)–(G_E3.3) hold. Choose an infinite κ such that if $\chi \not\models_{L_i}^* \chi'$, then for every $\kappa' \geq \kappa$ there exists an (E, κ') -homogeneous model $\mathfrak{M}_i$ based on an L_i -frame with $\mathfrak{M}_i \models \chi$ and $\mathfrak{M}_i \not\models \chi'$, for $i = 1, 2$. By choosing $\kappa' \geq \kappa$ sufficiently large and taking appropriate disjoint unions of models, it follows from (G_E3.1)–(G_E3.3) that, for each $i = 1, 2$, we can find an (E_i, κ') -homogeneous model $\mathfrak{M}_i = (\mathfrak{F}_i, V_i)$ with $\mathfrak{F}_i = (W_i, R_i, E_i)$ such that $\mathfrak{F}_i \models L_i$ and the following conditions hold:

1. $\mathfrak{M}_1 \models \varphi^1 \wedge \hat{\mathcal{Q}}^1$, $\mathfrak{M}_1 \not\models \psi^1$, and $|\{[w]_{E_1} \mid \mathfrak{M}_1, w \models \hat{q}^1\}| = \kappa'$ for all $q \in \mathcal{Q}$;
2. $\mathfrak{M}_2 \models \hat{\mathcal{Q}}^2$ and $|\{[w]_{E_2} \mid \mathfrak{M}_2, w \models \hat{q}^2\}| = \kappa'$ for all $q \in \mathcal{Q}$.

It follows that we find a bijection $F: W_1 \rightarrow W_2$ such that, for all $w \in W_1$:

1. there is a (uniquely determined) Φ -type t such that $\mathfrak{M}_1, w \models t^1$ and $\mathfrak{M}_2, f(w) \models t^2$;
2. $[f(w)]_{E_2} = \{f(w') \mid w' \in [w]_{E_1}\}$.

Now construct a new model $\mathfrak{M} = (\mathfrak{F}, V)$ by taking $\mathfrak{M}_1$, interpreting $\Box_2$ by $f^{-1}[R_2] = \{(w, w') \in W_1 \mid (f(w), f(w')) \in R_2\}$, and setting $V(p) = f^{-1}(V_2(p))$ for all variables p introduced as surrogates of subformulas of the form $\Box_2 \chi$. Then, for $\mathfrak{F} = (W_1, R_1, f^{-1}[R_2], E_1)$, we have $\mathfrak{F} \models L_1 \otimes_E L_2$ by definition, and one can show by induction for all formulas $\chi \in \Phi$ and $w \in W_1$:

$$\mathfrak{M}, w \models \chi \quad \text{iff} \quad \mathfrak{M}_1, w \models \chi^1 \quad \text{iff} \quad \mathfrak{M}_2, f(w) \models \chi^2.$$

It follows that $\mathfrak{M} \models \varphi$ and $\mathfrak{M} \not\models \psi$, as required. $\square$

The following lemma shows that the conditions of Theorem 5.2 are satisfied for (globally) Kripke complete (semi)commutators with **S5**.

Lemma 5.4. *If a logic of the form $[L, \mathbf{S5}]^{xd}$ or $[L, \mathbf{S5}]^{cd}$ is (globally) Kripke complete, then it admits E -homogeneous models (for global consequence).*

Proof. We give the proof for the local case, the proof in the global case is the same. Assume first that $[L, \mathbf{S5}]^{xd}$ is Kripke complete. Let κ be any infinite cardinal such that, for every $\varphi \notin [L, \mathbf{S5}]^{xd}$, there exists a model refuting φ based on an $[L, \mathbf{S5}]^{xd}$ -frame of size at most κ . We show that κ is as required. Assume $\kappa' \geq \kappa$ and $\varphi \notin [L, \mathbf{S5}]^{xd}$. By definition of κ , there exists a model $\mathfrak{M} = (\mathfrak{F}, V)$ with $\mathfrak{F} = (W, R, E)$, $\mathfrak{M} \not\models \varphi$, $\mathfrak{F} \models [L, \mathbf{S5}]^{xd}$, and W of size κ . Then define $\mathfrak{M}_{\kappa'} = (\mathfrak{F}_{\kappa'}, V_{\kappa'})$ with $\mathfrak{F}_{\kappa'} = (W', R', E')$ by setting

- $W' = W \times \kappa'$;
- $(w, n)R'(w', n')$ if wRw' and $n = n'$, for all $n, n' \in \kappa'$;
- $(w, n)E'(w', n')$ if wEw' , for all $n, n' \in \kappa'$;
- $V_{\kappa'}(p) = V(p) \times \kappa'$, for all proposition letters p .

Claim. $\mathfrak{F}_{\kappa'} \models [L, \mathbf{S5}]^{xd}$, $\mathfrak{M}_{\kappa'} \not\models \varphi$, and $\mathfrak{M}_{\kappa'}$ is an (E, κ') -homogeneous model.

Proof of Claim. By closure under disjoint unions, (W', R') validates L . Also, E' is easily seen to be an equivalence relation. We next show that (W', R', E') satisfies $(lcom_{R,E})$. Assume $(w, n)E'(w', m)$ and $(w', m)R'(w'', k)$. By definition, wEw' , $m = k$, and $w'Rw''$. Then, as (W, R, E) satisfies $(lcom_{R,E})$, we have a v' with wRv' and $v'Ew''$. But then $(w, n)R'(v', n)$ and $(v', n)E'(w'', k)$, as required.

Let χ be any formula. Then one can easily show by induction that $\mathfrak{M}, w \models \chi$ iff $\mathfrak{M}_{\kappa'}, (w, n) \models \chi$ for all $n \in \kappa'$. It follows that $\mathfrak{M}_{\kappa'} \not\models \varphi$ and $\mathfrak{M}_{\kappa'}$ is an (E, κ') -homogeneous model. This finishes the proof of the claim and also the proof for logics of the form $[L, \mathbf{S5}]^{xd}$.

Now assume a logic of the form $[L, \mathbf{S5}]^{cd}$ is given and Kripke complete. The construction of $\mathfrak{M}_{\kappa'}$ is the same, using a model $\mathfrak{M} = (\mathfrak{F}, V)$ with $\mathfrak{F}$ also satisfying $(rcom_{R,E})$. So it suffices to show that $\mathfrak{M}_{\kappa'}$ then also satisfies $(rcom_{R,E})$: assume $(w, n)R'(w', m)$ and $(w', m)E'(w'', k)$. By definition, $n = m$, wRw' , and $w'Ew''$. Then, as (W, R, E) satisfies $(rcom_{R,E})$, we have a v' with wEv' and $v'Rw''$. But then $(w, n)E'(v', k)$ and $(v', k)R'(w'', k)$, as required. $\square$

We obtain the following transfer result for (semi)commutators.

Theorem 5.5. *Let $d \in \{xd, cd\}$ and L_1 and L_2 be consistent propositional modal logics such that $[L_i, \mathbf{S5}]^d$ is (globally) Kripke complete, for both $i = 1, 2$. Then*

- $[L_1, \mathbf{S5}]^d \otimes_E [L_2, \mathbf{S5}]^d$ is (globally) Kripke complete;
- $[L_1, \mathbf{S5}]^d \otimes_E [L_2, \mathbf{S5}]^d$ is (globally) decidable iff $[L_1, \mathbf{S5}]^d$ and $[L_2, \mathbf{S5}]^d$ are (globally) decidable.

6 Discussion

We have established preservation and non-preservation results for one-variable first-order modal logics and for related propositional modal logics. Many questions remain open:

- The behaviour of fusions of full first-order modal logic is still largely unexplored. A transfer theorem for canonicity is given in [30], but, for example, the transfer of completeness is open.
- Monodic fragments extend one-variable modal logics by allowing more expressive first-order fragments while restricting modal operators to formulas with at most one free variable [25, 1]. We have shown that decidability does not transfer from the one-variable fragment with equality to the monodic fragment over the two-variable fragment of FO with counting. Positive preservation results are known [13], and it is straightforward to adapt our proofs to, for instance, the monodic fragments based on the two-variable fragment of FO without equality or description logics such as *ALC*, *ALCI*, and *SHI* with the universal role [3, 25]. However, a systematic study of fusions of monodic fragments is still missing.
- It would be valuable to identify sufficient conditions under which Kripke completeness and decidability transfer to fusions of one-variable modal logics with equality. We note that many standard polymodal fusions with equality such as polymodal **K** and **S5** are in fact decidable [1].
- We have seen that, in many cases, completeness and decidability are preserved under fusions of propositional modal logics sharing an **S5** operator. Can this be extended to other modal logics?
- Our analysis focused on Kripke complete modal logics. Since first-order modal logics are often incomplete under standard Kripke semantics, it would be interesting to obtain preservation results that do not presuppose Kripke completeness. Techniques from [33, 2] may be helpful here.

References

- [1] Alessandro Artale, Christopher Hampson, Roman Kontchakov, Andrea Mazzullo & Frank Wolter (2025): *Decidability in First-Order Modal Logic with Non-Rigid Constants and Definite Descriptions*. CoRR abs/2509.08165, doi:10.48550/arXiv.2509.08165. arXiv:2509.08165.
- [2] Franz Baader, Silvio Ghilardi & Cesare Tinelli (2006): *A new combination procedure for the word problem that generalizes fusion decidability results in modal logics*. *Inf. Comput.* 204(10), pp. 1413–1452, doi:10.1016/J.IC.2005.05.009.
- [3] Franz Baader, Ian Horrocks, Carsten Lutz & Ulrike Sattler (2017): *An Introduction to Description Logic*. Cambridge University Press, doi:10.1017/9781139025355.
- [4] Franz Baader, Carsten Lutz, Holger Sturm & Frank Wolter (2002): *Fusions of Description Logics and Abstract Description Systems*. *J. Artif. Intell. Res.* 16, pp. 1–58, doi:10.1613/JAIR.919.
- [5] Guram Bezhanishvili & Mashiath Khan (2024): *The Monadic Grzegorzczuk Logic*. CoRR abs/2412.10854, doi:10.48550/arXiv.2412.10854. arXiv:2412.10854.
- [6] Fredrik Dahlqvist & Dirk Pattinson (2011): *On the Fusion of Coalgebraic Logics*. In: *Proc. of the 4th Int. Conf. on Algebra and Coalgebra in Computer Science (CALCO 2011)*, *Lecture Notes in Computer Science* 6859, Springer, pp. 161–175, doi:10.1007/978-3-642-22944-2_12.
- [7] Martin Davis (1977): *Unsolvability Problems*. In Jon Barwise, editor: *Handbook of Mathematical Logic*, North-Holland, Amsterdam, pp. 567–594, doi:10.1016/S0049-237X(08)71115-7.
- [8] Anatoli Degtyarev, Michael Fisher & Alexei Lisitsa (2002): *Equality and Monodic First-Order Temporal Logic*. *Stud Logica* 72(2), pp. 147–156, doi:10.1023/A:1021352309671.
- [9] Kit Fine & Gerhard Schurz (1996): *Transfer Theorems for Multimodal Logics*. In B. Jack Copeland, editor: *Logic and Reality: Essays on the Legacy of Arthur Prior*, Oxford University Press, pp. 169–213.
- [10] Dov M. Gabbay (1996): *Fibred Semantics and the Weaving of Logics, Part 1: Modal and Intuitionistic Logics*. *J. Symb. Log.* 61(4), pp. 1057–1120, doi:10.2307/2275807.
- [11] Dov M. Gabbay & Valentin B. Shehtman (1998): *Products of Modal Logics, Part 1*. *Log. J. IGPL* 6(1), pp. 73–146, doi:10.1093/JIGPAL/6.1.73.
- [12] David Gabelaia, Agi Kurucz, Frank Wolter & Michael Zakharyashev (2006): *Non-primitive recursive decidability of products of modal logics with expanding domains*. *Ann. Pure Appl. Log.* 142(1-3), pp. 245–268, doi:10.1016/J.APAL.2006.01.001.
- [13] Silvio Ghilardi, Enrica Nicolini & Daniele Zucchelli (2008): *A comprehensive combination framework*. *ACM Trans. Comput. Log.* 9(2), pp. 8:1–8:54, doi:10.1145/1342991.1342992.
- [14] Silvio Ghilardi & Luigi Santocanale (2003): *Algebraic and Model Theoretic Techniques for Fusion Decidability in Modal Logics*. In: *Proc. of the 10th Int. Conf. on Logic for Programming, Artificial Intelligence, and Reasoning (LPAR 2003)*, *Lecture Notes in Computer Science* 2850, Springer, pp. 152–166, doi:10.1007/978-3-540-39813-4_10.
- [15] Robert Goldblatt (1987): *Logics of time and computation*. Center for the Study of Language and Information.
- [16] Valentin Goranko & Solomon Passy (1992): *Using the Universal Modality: Gains and Questions*. *J. Log. Comput.* 2(1), pp. 5–30, doi:10.1093/LOGCOM/2.1.5.
- [17] Christopher Hampson (2016): *Two-dimensional modal logics with difference relations*. Ph.D. thesis, King’s College London, UK. Available at <https://kclpure.kcl.ac.uk/portal/en/studentTheses/two-dimensional-modal-logics-with-difference-relations/>.
- [18] Christopher Hampson (2018): *The Bimodal Logic of Commuting Difference Operators Is Decidable*. In: *Proc. of the 12th Conf. on Advances in Modal Logic (AiML 2018)*, College Publications, pp. 311–326. Available at <http://www.aiml.net/volumes/volume12/Hampson.pdf>.

- [19] Christopher Hampson & Agi Kurucz (2015): *Undecidable Propositional Bimodal Logics and One-Variable First-Order Linear Temporal Logics with Counting*. *ACM Trans. Comput. Log.* 16(3), pp. 27:1–27:36, doi:10.1145/2757285.
- [20] Roman Kontchakov, Dmitry Shkatov & Frank Wolter (2026): *Fusions of One-Variable First-Order Modal Logics (Extended Version)*. CoRR abs/2603.04512, doi:10.48550/arXiv.2603.04512. arXiv:2603.04512.
- [21] Marcus Kracht & Frank Wolter (1991): *Properties of Independently Axiomatizable Bimodal Logics*. *J. Symb. Log.* 56(4), pp. 1469–1485, doi:10.2307/2275487.
- [22] Marcus Kracht & Frank Wolter (1999): *Normal Monomodal Logics Can Simulate All Others*. *J. Symb. Log.* 64(1), pp. 99–138, doi:10.2307/2586754.
- [23] Agi Kurucz (2007): *Combining Modal Logics*. In Patrick Blackburn, Johan van Benthem & Frank Wolter, editors: *Handbook of Modal Logic*, Elsevier, pp. 869–924, doi:10.1016/S1570-2464(07)80018-8.
- [24] Agi Kurucz, Frank Wolter & Michael Zakharyashev (2025): *Deciding the Existence of Interpolants and Definitions in First-Order Modal Logic*. *Log. Methods Comput. Sci.* 21(4), doi:10.46298/LMCS-21(4:6)2025.
- [25] Agi Kurucz, Frank Wolter, Michael Zakharyashev & Dov M. Gabbay (2003): *Many-Dimensional Modal Logics: Theory and Applications*. *Studies in Logic and the Foundations of Mathematics* 148, North Holland, Amsterdam.
- [26] Ian Pratt-Hartmann (2005): *Complexity of the Two-Variable Fragment with Counting Quantifiers*. *J. Log. Lang. Inf.* 14(3), pp. 369–395, doi:10.1007/S10849-005-5791-1.
- [27] Maarten de Rijke (1992): *The Modal Logic of Inequality*. *J. Symb. Log.* 57(2), pp. 566–584, doi:10.2307/2275293.
- [28] Valentin Shehtman & Dmitry Shkatov (2019): *On one-variable fragments of modal predicate logics*. In: *Proc. of SYMICS 2019, ILLC*, University of Amsterdam, pp. 129–132.
- [29] Valentin Shehtman & Dmitry Shkatov (2023): *Semiproducts, Products, and Modal Predicate Logics: Some Examples*. *Doklady. Mathematics* 108(2), pp. 411–418, doi:10.1134/S1064562423701296.
- [30] Valentin Shehtman & Dmitry Shkatov (2024): *Fusions of Canonical Predicate Modal Logics Are Canonical*. *Advances in Modal Logic* 2024, *Short Papers*, pp. 51–56.
- [31] Stephen K. Thomason (1980): *Independent Propositional Modal Logics*. *Studia Logica* 39, pp. 143–144, doi:10.1007/BF00370317.
- [32] M. Wajsberg (1933): *Ein erweiterter Klassenkalkül*. *Monatshefte für Mathematik und Physik* 40, pp. 113–126, doi:10.1007/BF01708856.
- [33] Frank Wolter (1996): *Fusions of Modal Logics Revisited*. In: *Proc. of the 1st Workshop on Advances in Modal Logic (AiML 1996)*, CSLI Publications, pp. 361–379.
- [34] Georg Henrik von Wright (1979): *A Modal Logic of Place*. In Ernest Sosa, editor: *The Philosophy of Nicholas Rescher*, D. Reidel, Dordrecht, pp. 65–73, doi:10.1007/978-94-009-9407-2_9.
- [35] Alberto Zanardo, Amílcar Sernadas & Cristina Sernadas (2001): *Fibering: Completeness Preservation*. *J. Symb. Log.* 66(1), pp. 414–439, doi:10.2307/2694931.